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Estimation and inference in spatial models with dominant units☆

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ABSTRACT

In spatial econometrics literature estimation and inference are carried out assuming that the matrix of spatial or network connections has uniformly bounded absolute column sums in the number of units, n, in the network. This paper relaxes this restriction and allows for one or more units to have pervasive effects in the network. The linear–quadratic central limit theorem of Kelejian and Prucha (2001) is generalized to allow for such dominant units, and the asymptotic properties of the GMM estimators are established in this more general setting. A new bias-corrected method of moments (BMM) estimator is also proposed that avoids the problem of weak instruments by self-instrumenting the spatially lagged dependent variable. Both cases of homoskedastic and heteroskedastic errors are considered and the associated estimators are shown to be consistent and asymptotically normal, depending on the rate at which the maximum column sum of the weights matrix rises with n. The small sample properties of GMM and BMM estimators are investigated by Monte Carlo experiments and shown to be satisfactory. An empirical application to sectoral price changes in the US over the pre- and post-2008 financial crisis is also provided. It is shown that the share of capital can be estimated reasonably well from the degree of sectoral interdependence using the input–output tables, despite the evidence of dominant sectors being present in the US economy.

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1. Introduction

In spatial econometrics, the interdependence among cross-sectional units is captured via a spatial weights matrix, $\mathbf{W} = (w_{ij})$, which is usually constructed based on some measures of geographical, economic or social distance. A critical assumption that has been adopted in the existing literature is that the maximum absolute row and column sum norms of $\mathbf{W}$ are uniformly bounded in the number of cross section units, n. This assumption, which dates back to the seminal contributions of Kelejian and Prucha (1998, 1999), essentially imposes a strong restriction on the degree of cross-sectional dependence amongst the units in the spatial model or network. It will be satisfied, for example, if $\mathbf{W}$ is sparse in the

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sense that each unit has only a finite number of "neighbors", or if the strength of their connections decays sufficiently fast with their distance from one another. However, such sparsity conditions rule out the possibility that some units could be dominant or influential, in the sense that they might impact a large number of other units in the network. This could arise, for example, in the case of production or financial networks where a large number of firms or households could depend on one or more banks or sectors in the economy, as documented in the recent contributions by Acemoglu et al. (2012), Dungey and Volkov (2018) and Pesaran and Yang (2020). Acemoglu et al. (2012) show that in the US intersectoral network there exists a high degree of asymmetry in the roles that sectors play as suppliers to others, and such asymmetries in production networks could be an important source of aggregate fluctuations. Pesaran and Yang (2020) confirm the finding in Acemoglu et al. (2012) and further identify the wholesale trade sector as the dominant sector in the US over the period 1972–2007. Dungey and Volkov (2018) consider 49 OECD economies from 1996 to 2011 and identify wholesale trade and R&D as two dominant sectors to most economies. In such cases the standard proofs used to justify the consistency and asymptotic normality of the proposed estimators are no longer applicable.

In this paper we consider estimation and inference in spatial autoregressive (SAR) models where the maximum column sum norm of the weights matrix, denoted by $\|\mathbf{W}\|_1$, is allowed to rise with the dimension of the network, n. Specifically, we suppose $\|\mathbf{W}\|_1 = \ominus\left(n^{\delta}\right)$, where $\ominus\left(n^{\delta}\right)$ denotes the expansion rate of $\|\mathbf{W}\|_1$ in terms of n, with $\delta \in [0, 1)$. The exponent δ measures the degree to which the most influential unit in the network impacts all other units. The condition imposed on $\mathbf{W}$ in the literature corresponds to assuming $\delta = 0$. But, as noted above, in many applications it is likely that $\delta > 0$, and it is therefore desirable to provide conditions under which standard estimators of SAR models continue to apply in such cases.

The exponent δ also relates to measures of network centrality. In the case of spatial models with row normalized weights matrices, the degree of centrality of unit j is typically measured by its (weighted) outdegree, defined by $d_{jn} = \sum_{i=1}^{n} w_{ij}$. The degree of dominance of unit j can now be measured by the exponent δ_j, defined by $d_{jn} = \ominus\left(n^{\delta_j}\right)$, where $\delta_j \in [0, 1]$. Unit j is said to be strongly dominant if $\delta_j = 1$, weakly dominant if $\delta_j > 0$, and non-dominant if $\delta_j = 0$.[1] To simplify the exposition we refer to unit j as being dominant if $\delta_j > 0$, unless it is important to distinguish between cases of strong and weak dominance. Accordingly, the overall degree of network centrality is also given by $\delta = \max(\delta_1, \delta_2, \ldots, \delta_n)$.[2] From this perspective, the assumption that $\mathbf{W}$ has bounded column sum norm requires that $\delta_j = 0$, for all j. The present paper relaxes this assumption and develops new estimation and inference theory allowing for the existence of dominant units ($\delta > 0$) in the network.[3]

We begin by generalizing the central limit theorem for linear–quadratic forms due to Kelejian and Prucha (2001), which requires $\delta = 0$. For our analysis we need to relax this restriction and allow the matrix in the quadratic form of their theorem to have column sums that are unbounded in n (namely allow for $\delta > 0$). The generalized central limit theorem is then used to establish the asymptotic properties of the estimators of the SAR model.

There are two main approaches to the estimation of spatial models, namely the quasi maximum likelihood (QML) method advanced by Cliff and Ord (1973, 1981), (Upton and Fingleton, 1985), and developed further by Anselin (1988), Lee (2004), Lee and Yu (2010), amongst others. The second approach is the generalized method of moments (GMM) pioneered by Kelejian and Prucha (1998, 1999), and extended and further studied by Lee (2007), Kapoor et al. (2007) , Lin and Lee (2010), and Lee and Yu (2014), amongst others. In this paper we consider the asymptotic properties of the GMM estimators of the SAR model under both homoskedastic and heteroskedastic errors, thus generalizing the results developed by Lee (2007) and Lin and Lee (2010) to the case of non-bounded spatial weights matrices. We establish conditions under which GMM estimators are consistent and asymptotically normal even if $\delta > 0$, under both homoskedastic and heteroskedastic errors.

We also propose a new bias-corrected method of moments (BMM), which is applicable generally and is simple to implement. The BMM approach was first introduced in a recent paper by Chudik and Pesaran (2020) for the estimation of dynamic panel data models with short time-dimension. In the context of the SAR model, the spatial lag variable is endogenous. Instead of looking for valid instruments, the BMM approach uses the spatial lag variable as an "instrument" for itself, but corrects the bias due to the non-zero correlation between the spatial lag variable and the error term. This method has the advantage of avoiding the weak instrument problem by design. We show that both GMM and BMM estimators are consistent if $0 \leq \delta < 1$, and establish their asymptotic normality for values of δ in the range $0 \leq \delta < 1/2$, irrespective of whether the errors are homoskedastic or not. A similar correction is also considered by Liu and Yang (2015) who propose modifying the score function of the log-likelihood function of a SAR model to ensure consistency of the quasi-maximum likelihood estimator when the pattern of heteroskedasticity is correlated with the spatial weights.

[1] For further details see Definition 1 in Pesaran and Yang (2020).

[2] Note that when $w_{ij} \geq 0$, we have $\|\mathbf{W}\|_1 = \sup_j(d_j)$.

[3] It is worth noting that in the current paper we assume $\mathbf{W}$ is known and focus on estimating the spatial parameters. In cases where information on direct connections of the network is unavailable, there exists a related literature that uses large panel data sets (with both n and T large) to detect which unit has the largest δ (when δ equals or is close to unity) from the pattern of correlation in the data without needing to know $\mathbf{W}$. See, for example, Parker and Sul (2016), Brownlees and Mesters (2020), and Kapetanios et al. (2020). In a related literature, Bailey et al. (2016b) also consider estimating δ using large panel data sets when $\mathbf{W}$ is not known.

The GMM and BMM estimators that we propose allow for such correlations, and are consistent under general forms of heteroskedastic errors.[4]

An extensive set of Monte Carlo experiments lend support to the theoretical results and document that both estimators have satisfactory small sample properties, with the BMM estimator outperforming the GMM estimator when n is relatively small and δ is close to unity. The estimation techniques are shown to be robust to different degrees of spatial dependence, various specifications of the spatial weights matrix, and non-Gaussian errors. Both estimators also perform reasonably well under general error heteroskedasticity.

As an empirical application we consider the sectoral price changes in the US over the pre- and post-2008 financial crisis, using 300 × 300 input–output tables as spatial weights. We show that the share of capital can be estimated from the degree of sectoral interdependence. We first investigate the presence of dominant sectors in the US economy by computing the extremum estimator of δ (the degree of network centrality) proposed in Pesaran and Yang (2020), and obtain estimates lying between 0.71 and 0.85, suggesting the existence of at least one dominant sector in the US economy. We then estimate a SAR model with homoskedastic errors in the rate of sectoral price changes and provide estimates of the share of capital of around 0.4 during the pre-crisis period (1998–2006), and 0.3 over the post-crisis period (2007–2015). We obtain slightly larger estimates for the pre-crisis period when we allow for heteroskedastic errors, which suggests error heteroskedasticity might have been serious in the pre-crisis period. But overall, the estimates of capital share obtained using the SAR model compare reasonably well with the estimates reported in the literature using very different calibration techniques.

The remainder of the paper is organized as follows: Section 2 describes the model and sets out its assumptions. Section 3 provides a generalization of Kelejian and Prucha's central limit theorem. The GMM and BMM estimation methods and their asymptotic properties are detailed in Sections 4 and 5, respectively. Section 6 presents the finite sample properties of the GMM and BMM estimators using Monte Carlo techniques. Section 7 contains the empirical application, and Section 8 gives some concluding remarks. An Online Supplement (https://doi.org/10.1016/j.jeconom.2020.04.045) provides proofs of theorems and propositions, gives statements and proofs of necessary lemmas, and includes additional Monte Carlo and empirical results.

Notations. Generic positive finite constants are denoted by K when they are large, and by c when small. They can take different values at different instances. Let $\{f_n\}_{n=1}^{\infty}$ be a real sequence and $\{g_n\}_{n=1}^{\infty}$ be a real positive sequence. We write $f_n = O(g_n)$ if there exists a positive finite constant K_0 such that $|f_n|/g_n \leq K_0$ for all n; we write $f_n = o(g_n)$ if $f_n/g_n \to 0$ as $n \to \infty$. If $\{f_n\}_{n=1}^{\infty}$ and $\{g_n\}_{n=1}^{\infty}$ are both positive sequences of real numbers, then $f_n = \ominus(g_n)$ if there exist $N_0 \geq 1$ and positive finite constants K_0 and K_1, such that $\inf_{n \geq N_0}(f_n/g_n) \geq K_0$, and $\sup_{n \geq N_0}(f_n/g_n) \leq K_1$. The symbols $\to_p$ and $\to_d$ indicate convergence in probability and in distribution as $n \to \infty$, respectively. Let $\{x_n\}$ be a sequence of random variables. We write $x_n = o_p(1)$ if $x_n \to_p 0$ as $n \to \infty$. $E_0(\cdot)$ denotes expectations taken under the true probability measure. For an $n \times n$ matrix $\mathbf{A} = (a_{ij})$, $\|\mathbf{A}\|_\infty = \sup_{1 \leq i \leq n} \sum_{j=1}^{n} |a_{ij}|$ denotes the maximum absolute row sum norm (or row norm, for short) of $\mathbf{A}$; $\|\mathbf{A}\|_1 = \sup_{1 \leq j \leq n} \sum_{i=1}^{n} |a_{ij}|$ denotes the maximum absolute column sum norm (or column norm); and $\lambda_{\max}(\mathbf{A})$ ($\lambda_{\min}(\mathbf{A})$) denotes the largest (smallest) eigenvalue of $\mathbf{A}$. The symbol *diag* ($\mathbf{A}$) represents a *vector* consisting of the diagonal elements of $\mathbf{A}$, namely, *diag* $(\mathbf{A}) = (a_{11}, a_{22}, \ldots, a_{nn})'$; whereas *Diag* ($\mathbf{A}$) represents a *diagonal matrix* formed by the diagonal entries of $\mathbf{A}$. $\mathbf{1}_n$ denotes the $n \times 1$ vector of ones.

2. The model and its assumptions

We consider the following standard SAR model:

$$y_i = \rho y_i^* + \boldsymbol{\beta}' \mathbf{x}_i + \varepsilon_i, \text{ for } i = 1, 2, \ldots, n, \tag{1}$$

where y_i is the outcome variable on unit i, ρ is a fixed spatial coefficient, $\mathbf{x}_i$ is a $k \times 1$ vector of regressors on unit i with the associated vector of fixed coefficients $\boldsymbol{\beta}$, ε_i is a random error, y_i^* is the spatial variable, defined by

$$y_i^* = \sum_{j=1}^{n} w_{ij} y_j = \mathbf{w}_{i.}' \mathbf{y}, \tag{2}$$

$\mathbf{y} = (y_1, y_2, \ldots, y_n)'$, $\mathbf{w}_{i.} = (w_{i1}, w_{i2}, \ldots, w_{in})'$ is a vector of known constant weights and $w_{ij} \geq 0$ for all i and j. Let $\mathbf{y}^* = (y_1^*, y_2^*, \ldots, y_n^*)'$. Then (2) implies that $\mathbf{y}^* = \mathbf{W}\mathbf{y}$, where $\mathbf{W} = (w_{ij}) = (\mathbf{w}_{1.}, \mathbf{w}_{2.}, \ldots, \mathbf{w}_{n.})'$ is an $n \times n$ known matrix of spatial weights (or network connections).

We suppose that the row sums of $\mathbf{W}$ are uniformly bounded in n, but allow the column sums of $\mathbf{W}$ to rise with n. Specifically, following Pesaran and Yang (2020), denote the jth column sum of $\mathbf{W}$ by $d_{jn} = \sum_{i=1}^{n} w_{ij}$, and assume that d_{jn} is of order n^{δ_j} such that

$$d_{jn} = \kappa_j n^{\delta_j}, \text{ for } j = 1, 2, \ldots, n, \tag{3}$$

[4] It is also worth noting that by correcting the score function Liu and Yang (2015) end up with a variance estimator which is quite complicated and involves higher order moments of the errors, unlike the BMM estimator whose covariance matrix is relatively simple and does not depend on the higher order moments. Furthermore, Liu and Yang (2015) conduct their analysis under the standard assumption that the spatial weights matrix is uniformly bounded in both row and column norms.

where δ_j is a fixed constant in the range $0 \leq \delta_j \leq 1$, and κ_j is a strictly positive random variable defined on $0 < \underline{\kappa} \leq \kappa_j \leq \bar{\kappa} < K$, where $\underline{\kappa}$ and $\bar{\kappa}$ are fixed constants. We also set

$$\delta = \max_{j=1,2,\ldots,n} \left(\delta_j\right), \quad 0 \leq \delta \leq 1, \tag{4}$$

and note that $\max_j \left(d_{jn}\right) = \|\mathbf{W}\|_1 = \ominus(n^{\delta})$. We further assume that the number of dominant units, m, is finite, and without loss of generality we suppose the first m units, $j = 1, 2, \ldots, m$, are δ_j-dominant (with $\delta_j > 0$), and the rest of the units, $j = m+1, m+2, \ldots, n$, are non-dominant (with $\delta_j = 0$). In particular, the spatial weights matrix for the non-dominant units is denoted by $\mathbf{W}_{22}$, which is the $(n-m)$-dimensional square submatrix of $\mathbf{W}$ that captures the connections among the non-dominant units. In short, we consider weights matrices such that their first m column sums are unbounded in n, with the remaining $(n-m)$ column sums bounded. Although m is assumed to be fixed, it can be shown that this must be true if δ_j's satisfy the summability condition, $\sum_{j=1}^{n} \delta_j < K$.[5] Also, since our focus is on the estimation of ρ and $\boldsymbol{\beta}$, the identity of the dominant units or the order by which they are included in $\mathbf{W}$ does not affect the analysis.

In matrix notation, model (1) can be rewritten as

$$\mathbf{y} = \rho \mathbf{y}^* + \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}, \tag{5}$$

where $\mathbf{X} = (\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_n)'$ is an $n \times k$ matrix of observations on exogenous regressors, and $\boldsymbol{\varepsilon} = (\varepsilon_1, \varepsilon_2, \ldots, \varepsilon_n)'$. The reduced-form representation of (5) is given by

$$\mathbf{y} = \mathbf{S}^{-1}(\rho)\left(\mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}\right), \tag{6}$$

where $\mathbf{S}(\rho) = \mathbf{I}_n - \rho\mathbf{W}$. The existence of $\mathbf{S}^{-1}(\rho)$ is ensured under the assumptions to be discussed below. It immediately follows from (6) that

$$\mathbf{y}^* = \mathbf{W}\mathbf{y} = \mathbf{W}\mathbf{S}^{-1}(\rho)\left(\mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}\right) = \mathbf{G}(\rho)\left(\mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}\right), \tag{7}$$

where $\mathbf{G}(\rho) = \mathbf{W}\mathbf{S}^{-1}(\rho)$. Note that the variables and spatial weights may depend on the sample size and form triangular arrays, although we suppress subscript n for notational simplicity.

The parameters of interest are ρ and $\boldsymbol{\beta}$, and their true values are denoted by ρ_0 and $\boldsymbol{\beta}_0$, respectively. For ease of exposition, we use $\mathbf{S}_0$ to denote the matrix $\mathbf{S}(\rho)$ evaluated at the true parameter value ρ_0, namely, $\mathbf{S}_0 = \mathbf{S}(\rho_0) = \mathbf{I}_n - \rho_0\mathbf{W}$. Similarly, we set

$$\mathbf{G}_0 = \mathbf{G}(\rho_0) = \mathbf{W}\left(\mathbf{I}_n - \rho_0\mathbf{W}\right)^{-1} = \mathbf{W}\mathbf{S}_0^{-1}, \text{ and } \boldsymbol{\eta}_0 = \mathbf{G}_0\mathbf{X}\boldsymbol{\beta}_0. \tag{8}$$

The following assumptions are made to carry out the asymptotic analysis.

Assumption 1. The idiosyncratic errors, ε_i, for $i = 1, 2, \ldots, n$, in the SAR model given by (1) are independently distributed over i with zero means and variances, σ_i^2, such that $\inf_i\left(\sigma_i^2\right) > c > 0$, $\sup_i\left(\sigma_i^2\right) < K$, and $\sup_i E\left|\varepsilon_i\right|^{4+c} < K$, for some $c > 0$.

Assumption 2. The $(k+1)$-dimensional parameter vector in (1), $\boldsymbol{\psi} = \left(\rho, \boldsymbol{\beta}'\right)' \in \boldsymbol{\Psi} = \boldsymbol{\Theta}_{\rho} \times \boldsymbol{\Theta}_{\beta}$, where $\boldsymbol{\Theta}_{\rho}$ and $\boldsymbol{\Theta}_{\beta}$ are compact subsets of $\mathbb{R}$ and $\mathbb{R}^k$, respectively ($|\rho| < K$ and $\|\boldsymbol{\beta}\|_1 < K$); the true value of $\boldsymbol{\psi}$, denoted by $\boldsymbol{\psi}_0 = \left(\rho_0, \boldsymbol{\beta}_0'\right)'$, lies in the interior of the parameter space, $\boldsymbol{\Psi}$.

Assumption 3. Let $\mathbf{X} = (\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_n)'$ be the $n \times k$ matrix of observations on the regressors in (1), where $\mathbf{x}_i = (x_{i1}, x_{i2}, \ldots, x_{ik})'$, for $i = 1, 2, \ldots, n$. (a) $\mathbf{x}_i$ are distributed independently of the errors, ε_j, for all i and j, and $\sup_{i,s} E\left(|x_{is}|^{2+c}\right) < K$, (b) $n^{-1}\mathbf{X}'\mathbf{X} \to_p \boldsymbol{\Sigma}_{xx}$ is positive definite, and (c) $n^{-1}\mathbf{X}'\mathbf{G}_0\mathbf{X} \to_p \boldsymbol{\Sigma}_{xg_0x}$ and $n^{-1}\mathbf{X}'\mathbf{G}_0'\mathbf{G}_0\mathbf{X} \to_p \boldsymbol{\Sigma}_{xg_0g_0x}$, where $\mathbf{G}_0$ is defined by (8), and $\boldsymbol{\Sigma}_{xg_0x}$ and $\boldsymbol{\Sigma}_{xg_0g_0x}$ are finite $k \times k$ matrices.

Assumption 4. The spatial weights matrix, $\mathbf{W} = \left(w_{ij}\right)$, in the SAR model given by (1) is such that (a) $w_{ij} \geq 0$ for all i and j, (b) $\|\mathbf{W}\|_{\infty} < K$, and $|\rho|\,\|\mathbf{W}\|_{\infty} < 1$, and (c) its column sums, denoted by $d_{jn} = \sum_{i=1}^{n} w_{ij}$, $j = 1, 2, \ldots, n$, are non-zero and follow the specification given by (3), where $0 < \delta_j \leq 1$ for $j = 1, 2, \ldots, m$, and $\delta_j = 0$ for $j = m+1, m+2, \ldots, n$, with m being a fixed number. Also, $|\rho|\,\|\mathbf{W}_{22}\|_1 < 1$, where $\mathbf{W}_{22}$ is the $(n-m)$-dimensional square submatrix of $\mathbf{W}$ that represents the connections among the non-dominant units.

Assumption 5. There exists n_0 such that for all $n \geq n_0$ (including $n \to \infty$), either
(a) $n^{-1}\mathbf{Q}_0'\mathbf{Q}_0$ is positive definite, where $\mathbf{Q}_0 = \left(\mathbf{G}_0\mathbf{X}\boldsymbol{\beta}_0, \mathbf{X}\right)$,
and/or

[5] See Proposition 2 of Pesaran and Yang (2020). The assumption of a fixed number of dominant units is analogous to the assumption of a fixed number of strong factors in factor models. As shown in Chudik et al. (2011), in order for the variances of the observables to be bounded, the number of strong factors must be fixed and cannot rise with n.

(b) $h_n > c > 0$, where

$$h_n = Tr\left[n^{-1}\left(\mathbf{G}_0^2 + \mathbf{G}_0'\mathbf{G}_0 - 2\check{\mathbf{G}}_0\mathbf{G}_0\right)\boldsymbol{\Sigma}_0\right], \tag{9}$$

$\mathbf{G}_0$ is given by (8), $\check{\mathbf{G}}_0 = Diag\,(\mathbf{G}_0)$, and $\boldsymbol{\Sigma}_0 = Diag\left(\sigma_{10}^2, \sigma_{20}^2, \ldots, \sigma_{n0}^2\right)$, with σ_{i0}^2 denoting the true value of $\sigma_i^2 = Var(\varepsilon_i)$, for $i = 1, 2, \ldots, n$.

Remark 1. It is worth noting that under Assumption 4, $|\lambda_{max}(\rho\mathbf{W})| \leq |\rho|\,\|\mathbf{W}\|_\infty < 1$, and hence $\lambda_{min}\,[\mathbf{S}(\rho)] = 1 - \lambda_{max}\,(\rho\mathbf{W}) > c > 0$, and as a result $\mathbf{S}\,(\rho) = \mathbf{I}_n - \rho\mathbf{W}$ is invertible for all ρ satisfying $|\rho|\,\|\mathbf{W}\|_\infty < 1$, irrespective of whether the column sums of $\mathbf{W}$ are bounded or not.[6] Moreover, it can be seen from (6) that in order for the variance of y_i to be bounded, we need $\left\|\mathbf{S}^{-1}\,(\rho)\right\|_\infty < K$, which is ensured by the assumption $|\rho|\,\|\mathbf{W}\|_\infty < 1$.[7] It is also clear that the condition $\|\mathbf{W}\|_\infty < K$ of Assumption 4 follows from $|\rho|\,\|\mathbf{W}\|_\infty < 1$ when $|\rho| > 0$, but it is required when $\rho = 0$. In the special case where $\mathbf{W}$ is row-standardized such that each row sums up to one, the invertibility condition reduces to $|\rho| < 1$. Finally, dominant units with $\delta_j > 0$ can exist only in non-symmetric networks. This follows since when $\mathbf{W}$ is symmetric the boundedness of $Var(y_i)$ excludes the possibility of dominant units.

Remark 2. The non-negativity condition, $w_{ij} \geq 0$, in Assumption 4(a) is imposed only for ease of exposition and is not restrictive. When it fails to hold, one can decompose $\mathbf{W}$ into two weights matrices with non-negative elements, namely, $\mathbf{W} = \mathbf{W}^+ - \mathbf{W}^- = (w_{ij}^+) - (w_{ij}^-)$, with w_{ij}^+ and $w_{ij}^- \geq 0$. Then model (1) can be written as $\mathbf{y} = \rho_1\mathbf{W}^+\mathbf{y} + \rho_2\mathbf{W}^-\mathbf{y} + \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$. See Bailey et al. (2016a) for an empirical application employing this strategy.

Remark 3. In Assumption 3(c), we have assumed that $n^{-1}\mathbf{X}'\mathbf{G}_0'\mathbf{G}_0\mathbf{X} \to_p \boldsymbol{\Sigma}_{xg_0g_0x}$. As is proved in Lemma S.9 of the Online Supplement, $E\left(n^{-1}\mathbf{X}'\mathbf{G}_0'\mathbf{G}_0\mathbf{X}\right)$ exists and is bounded. Intuitively, since $\|\mathbf{G}_0\|_\infty < K$ by Lemma S.6, asymptotically $\mathbf{G}_0\mathbf{X}$ behaves similarly to $\mathbf{X}$.

3. A generalization of the central limit theorem for linear–quadratic forms of Kelejian and Prucha (2001)

To allow for the presence of dominant units in the SAR model, we need to generalize the central limit theorem established in Theorem 1 of Kelejian and Prucha (2001) for linear–quadratic forms. We first consider the quadratic term which helps clarify the role played by the rate at which the column sum norm of the $n \times n$ weights matrix, $\mathbf{W}$, varies with n. We then consider the extension of this theorem to linear–quadratic forms needed for the analysis of SAR models with exogenous regressors. In what follows we state the theorems and relegate their proofs to Section S1.2 of the Online Supplement.[8]

Theorem 1. *Let $\boldsymbol{\varepsilon} = (\varepsilon_1, \varepsilon_2, \ldots, \varepsilon_n)'$ denote the $n \times 1$ vector of random variables, where ε_i, for $i = 1, 2, \ldots, n$, are independently distributed over i with zero means and variance σ_i^2, where $\inf_i\left(\sigma_i^2\right) > c > 0$ and $\sup_i\left(\sigma_i^2\right) < K$. Suppose that $\sup_i E\,|\varepsilon_i|^{4+c} < K$, for some $c > 0$, and denote the excess kurtosis of $\{\varepsilon_i\}$ by $k_{e,i} = \left(\mu_{4i}/\sigma_i^4\right) - 3$, where $\mu_{4i} = E\left(\varepsilon_i^4\right)$. Let $\mathbf{P} = \left(p_{ij}\right)$ be an array of $n \times n$ constant matrices that satisfy the following conditions*

$$\|\mathbf{P}\|_\infty = \sup_i \sum_{j=1}^n \left|p_{ij}\right| < K, \tag{10}$$

$$\|\mathbf{P}\|_1 = \sup_j \sum_{i=1}^n \left|p_{ij}\right| = O\,(n^\alpha), \quad 0 \leq \alpha < 1, \tag{11}$$

where α is a fixed constant and $\mathbf{P}$ has a finite number of unbounded columns. Define $\mathbf{A} = (a_{ij}) = \left(\mathbf{P} + \mathbf{P}'\right)/2$. Suppose $\mathbf{A}$ is such that

$$n^{-1}\sum_{i=1}^n\sum_{j=1}^n a_{ij}^2\sigma_i^2\sigma_j^2 + \frac{1}{2}n^{-1}\sum_{i=1}^n a_{ii}^2\sigma_i^4 k_{e,i} > c > 0, \text{ for all } n \text{ (including } n \to \infty). \tag{12}$$

Then if α lies in the range $0 \leq \alpha < 1/2$, we have

$$Q = \frac{\boldsymbol{\varepsilon}'\mathbf{A}\boldsymbol{\varepsilon} - \sum_{i=1}^n a_{ii}\sigma_i^2}{\sqrt{n}\varpi_n} \to_d N(0, 1), \text{ as } n \to \infty, \tag{13}$$

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[6] This can be seen by noting that $\mathbf{S}(\rho)$ is invertible if $|\lambda_{\max}\,(\rho\mathbf{W})| < 1$, and $|\lambda_{\max}\,(\rho\mathbf{W})| \leq |\rho|\min\left(\|\mathbf{W}\|_\infty, \|\mathbf{W}\|_1\right)$.

[7] Without loss of generality, let us abstract from the regressors and note that under the invertibility condition $\mathbf{y} = \mathbf{S}^{-1}\,(\rho)\,\boldsymbol{\varepsilon}$. It then readily follows that $\inf_j\left(\sigma_j^2\right)\sum_{j=1}^n\left(s^{ij}\right)^2 \leq Var(y_i) = \sum_{j=1}^n\left(s^{ij}\right)^2\sigma_j^2 \leq \sup_j\left(\sigma_j^2\right)\sum_{j=1}^n\left(s^{ij}\right)^2$, where s^{ij} is the (i,j)th element of $\mathbf{S}^{-1}\,(\rho)$. Accordingly, for $Var(y_i)$ to be bounded in n and strictly positive, under Assumption 1 it is necessary and sufficient that the rows of $\mathbf{S}^{-1}\,(\rho)$ are square summable. This condition is in turn met when $|\rho|\,\|\mathbf{W}\|_\infty < 1$ even if $\|\mathbf{W}\|_1 = \ominus(n^\delta)$, with $\delta > 0$.

[8] Note that the elements of the weights matrix, $\mathbf{W}$, and the error vector, $\boldsymbol{\varepsilon}$, typically depend on n, the sample size. But, unless required for clarity, we suppress subscript n to simplify the notations.

where

$$\varpi_n^2 = 2n^{-1}\sum_{i=1}^{n}\sum_{j=1}^{n} a_{ij}^2\sigma_i^2\sigma_j^2 + n^{-1}\sum_{i=1}^{n} a_{ii}^2\sigma_i^4 k_{e,i}. \tag{14}$$

In application of the above theorem to GMM and BMM estimators of ρ, the column norm properties of the weights matrix, $\mathbf{W}$, carry over to matrix $\mathbf{P}$ in the above theorem, and allow us to establish asymptotic normality of the estimators even if $\mathbf{W}$ has unbounded column norms. It is also worth noting that matrix $\mathbf{P}$ in the above theorem need not be row-standardized, and our results hold as long as $\mathbf{P}$ is uniformly bounded in row norms, as required by (10).

Remark 4. It is easily seen that condition (12) ensures $\varpi_n^2 > c > 0$, for all n (including $n \to \infty$). If the errors are normally distributed, then $k_{e,i} = 0$ for all i, and

$$n^{-1}\sum_{i=1}^{n}\sum_{j=1}^{n} a_{ij}^2\sigma_i^2\sigma_j^2 \geq \left(\inf_i \sigma_i^2\right)^2 n^{-1}\sum_{i=1}^{n}\sum_{j=1}^{n} a_{ij}^2 = \left(\inf_i \sigma_i^2\right)^2 n^{-1}Tr\left(\mathbf{A}\mathbf{A}'\right) = \left(\inf_i \sigma_i^2\right)^2 n^{-1}Tr\left(\mathbf{A}^2\right),$$

then (12) simplifies to $n^{-1}Tr\left(\mathbf{A}^2\right) > c > 0$, which always holds true for finite n (except for the trivial case of $\mathbf{A} = \mathbf{0}$). Therefore in the case of $k_{e,i} = 0$, to ensure $\varpi_n^2 > c > 0$, it is sufficient to assume that $n^{-1}Tr\left(\mathbf{A}^2\right)$ tends to a strictly positive limit as $n \to \infty$.

The next theorem extends Theorem 1 to linear–quadratic forms.

Theorem 2. *Let $\boldsymbol{\varepsilon} = (\varepsilon_1, \varepsilon_2, \ldots, \varepsilon_n)'$ denote the $n \times 1$ vector of random variables, where ε_i, for $i = 1, 2, \ldots, n$, are independently distributed over i with zero means and variance σ_i^2, where $\inf_i\left(\sigma_i^2\right) > c > 0$ and $\sup_i\left(\sigma_i^2\right) < K$. Suppose that $\sup_i E\left|\varepsilon_i\right|^{4+c} < K$, for some $c > 0$, and denote the excess kurtosis of $\{\varepsilon_i\}$ by $k_{e,i} = \left(\mu_{4i}/\sigma_i^4\right) - 3$, where $\mu_{4i} = E\left(\varepsilon_i^4\right)$. Let $\boldsymbol{\eta} = (\eta_1, \eta_2, \ldots, \eta_n)'$ be a vector of random variables with means $\mu_{\eta,i}$ and variances $\sigma_{\eta,i}^2$, distributed independently of ε_j, for all i and j, where $\sigma_{\eta,i}^2 > 0$, for all i, and $\sup_i E\left(\left|\eta_i\right|^{2+c}\right) < K$. Let $\mathbf{P} = (p_{ij})$ be an array of $n \times n$ constant matrices that satisfy conditions (10) and (11), and $\mathbf{P}$ has a finite number of unbounded columns, with $\alpha \geq 0$, as defined by (11). Let $\mathbf{A} = (a_{ij}) = (\mathbf{P} + \mathbf{P}')/2$, and suppose that*

$$n^{-1}\sum_{i=1}^{n}\sum_{j=1}^{n} a_{ij}^2\sigma_i^2\sigma_j^2 + \frac{1}{2}n^{-1}\sum_{i=1}^{n} a_{ii}^2\sigma_i^4 k_{e,i} + \frac{1}{2}n^{-1}\sum_{i=1}^{n}\sigma_{\eta,i}^2\sigma_i^2 + n^{-1}\sum_{i=1}^{n} a_{ii}\mu_{\eta,i}\mu_{3i} > c > 0, \tag{15}$$

for all n (including $n \to \infty$), where $\mu_{3i} = E\left(\varepsilon_i^3\right)$. Then if α lies in the range $0 \leq \alpha < 1/2$, we have

$$\tilde{Q} = \frac{\boldsymbol{\varepsilon}'\mathbf{A}\boldsymbol{\varepsilon} + \boldsymbol{\varepsilon}'\boldsymbol{\eta} - \sum_{i=1}^{n} a_{ii}\sigma_i^2}{\sqrt{n}\tilde{\varpi}_n} \to_d N(0,1), \text{ as } n \to \infty, \tag{16}$$

where

$$\tilde{\varpi}_n^2 = 2n^{-1}\sum_{i=1}^{n}\sum_{j=1}^{n} a_{ij}^2\sigma_i^2\sigma_j^2 + n^{-1}\sum_{i=1}^{n} a_{ii}^2\sigma_i^4 k_{e,i} + n^{-1}\sum_{i=1}^{n}\sigma_{\eta,i}^2\sigma_i^2 + 2n^{-1}\sum_{i=1}^{n} a_{ii}\mu_{\eta,i}\mu_{3i}. \tag{17}$$

Remark 5. Condition (15) ensures that $\tilde{\varpi}_n^2 > c > 0$, for all n (including $n \to \infty$). If the errors are symmetrically distributed, then $\mu_{3i} = 0$. Since $\sigma_{\eta,i}^2 > 0$ for all i, condition (15) in this case would reduce to (12) of Theorem 1.

4. GMM estimation

We begin by extending the GMM method proposed by Lee (2007) for standard SAR models to the case where the column sums of the spatial weights matrix are not necessarily bounded in n. Lee (2007) suggests using both linear moment conditions formed with instruments and additional quadratic moments that are based on the properties of the idiosyncratic errors. Lee (2007) assumes that the errors are homoskedastic, and Lin and Lee (2010) further consider the GMM method in the presence of unknown heteroskedasticity. To make our analysis more general, our theoretical derivations allow for unknown error heteroskedasticity, and we make comments on the special case of homoskedastic errors.

Specifically, consider model (1) and suppose that $\mathbf{Z} = (\mathbf{z}_1, \mathbf{z}_2, \ldots, \mathbf{z}_n)'$ is an $n \times r$ $(r \geq k+1)$ matrix of instruments for the regressors $(\mathbf{y}^*, \mathbf{X})$. Formally, $\mathbf{Z}$ satisfies the following assumption:

Assumption 6. Let $\mathbf{Z} = (\mathbf{z}_1, \mathbf{z}_2, \ldots, \mathbf{z}_n)'$ be the $n \times r$ matrix of observations on the r instrumental variables, $\mathbf{z}_i = (z_{i1}, z_{i2}, \ldots, z_{ir})'$. (a) $\mathbf{z}_i$ is distributed independently of the errors, ε_j, for all i and $j = 1, 2, \ldots, n$, and $\sup_{i,s} E\left(\left|z_{is}\right|^{2+c}\right) < K$, (b) $n^{-1}\mathbf{Z}'\mathbf{Z} \to_p \boldsymbol{\Sigma}_{zz}$, a positive definite matrix, and (c) $n^{-1}\mathbf{Z}'\mathbf{Q}_0 \to_p \boldsymbol{\Sigma}_{zq,0}$ is a full column rank matrix, where $\mathbf{Q}_0 = \left(\mathbf{G}_0\mathbf{X}\boldsymbol{\beta}_0, \mathbf{X}\right)$.

Recall that $\boldsymbol{\psi}$ denotes the $(k+1)$-dimensional vector of parameters, $\boldsymbol{\psi} = \left(\rho, \boldsymbol{\beta}'\right)'$ and its true value is denoted by $\boldsymbol{\psi}_0 = \left(\rho_0, \boldsymbol{\beta}_0'\right)'$. The r linear moment conditions are given by:

$$E_0\left[\mathbf{Z}'\boldsymbol{\varepsilon}\left(\boldsymbol{\psi}\right)\right] = \mathbf{0}, \tag{18}$$

where

$$\boldsymbol{\varepsilon}\left(\boldsymbol{\psi}\right) = \mathbf{y} - \rho\mathbf{y}^* - \mathbf{X}\boldsymbol{\beta}. \tag{19}$$

Since $\mathbf{X}$ is strictly exogenous under Assumption 3, a possible candidate for $\mathbf{Z}$ consists of linearly independent columns of $\left(\mathbf{X}, \mathbf{WX}, \mathbf{W}^2\mathbf{X}, \ldots\right)$. This choice of instruments was first proposed by Kelejian and Prucha (1998). To see why $\mathbf{Z}$ could take this form, note from (7) that $E\left(\mathbf{y}^*|\mathbf{X}\right) = \mathbf{G}\left(\rho\right)\mathbf{X}\boldsymbol{\beta}$. This term is clearly correlated with $\mathbf{y}^*$ but uncorrelated with $\boldsymbol{\varepsilon}$. Since $|\rho|\left\|\mathbf{W}\right\|_\infty < 1$ under Assumption 4(b), $\mathbf{G}\left(\rho\right)$ can be expanded as

$$\mathbf{G}\left(\rho\right) = \mathbf{W}\left(\mathbf{I}_n - \rho\mathbf{W}\right)^{-1} = \mathbf{W} + \rho\mathbf{W}^2 + \rho^2\mathbf{W}^3 + \cdots, \tag{20}$$

and then $\mathbf{G}\left(\rho\right)\mathbf{X}\boldsymbol{\beta} = \sum_{j=1}^{\infty}\rho^{j-1}\mathbf{W}^j\mathbf{X}\boldsymbol{\beta}$. This implies that the instruments for $\mathbf{y}^*$ can be chosen from the columns of $\left(\mathbf{WX}, \mathbf{W}^2\mathbf{X}, \ldots\right)$. Furthermore, Lee (2003) has shown that the asymptotically best IV matrix within the 2SLS framework is given by $\mathbf{Q}_0 = \left(\mathbf{G}_0\mathbf{X}\boldsymbol{\beta}_0, \mathbf{X}\right)$. Since $\mathbf{Q}_0$ depends on the unknown parameters ρ_0 and $\boldsymbol{\beta}_0$, a feasible best IV can be constructed using some initial consistent estimates of the parameters.

Turning to the quadratic moment condition, we recall that the idiosyncratic errors are assumed to be cross-sectionally uncorrelated. Using this property we have the following moment condition:

$$E_0\left[\boldsymbol{\varepsilon}'\left(\boldsymbol{\psi}\right)\mathbf{C}\boldsymbol{\varepsilon}\left(\boldsymbol{\psi}\right)\right] = 0, \tag{21}$$

where $\boldsymbol{\varepsilon}\left(\boldsymbol{\psi}\right)$ is defined by (19),

$$\mathbf{C} = \left(c_{ij}\right) = \left(\mathbf{B} + \mathbf{B}'\right)/2, \tag{22}$$

and $\mathbf{B}$ is a matrix that satisfies the following assumption:

Assumption 7. The matrix $\mathbf{B} = \left(b_{ij}\right)$ is an $n \times n$ matrix of fixed constants such that (a) $diag(\mathbf{B}) = \mathbf{0}$, (b) $\left\|\mathbf{B}\right\|_\infty < K$, (c) $\left\|\mathbf{B}\right\|_1 = O\left(n^{\delta_b}\right)$, where δ_b is a fixed constant in the range $0 \leq \delta_b < 1$, (d) $n^{-1}\mathbf{X}'\mathbf{CX} \to_p \boldsymbol{\Sigma}_{xcx}$, $n^{-1}\boldsymbol{\eta}_0'\mathbf{C}\boldsymbol{\eta}_0 \to_p c_0$, and $n^{-1}\mathbf{X}'\mathbf{C}\boldsymbol{\eta}_0 \to_p \mathbf{d}_0$, where $\boldsymbol{\eta}_0$ is given by (8), $\mathbf{C} = \left(\mathbf{B} + \mathbf{B}'\right)/2$, and $\mathbf{X}$ is the $n \times k$ matrix of observations on the regressors in model (1).

Let $\boldsymbol{\Sigma} = Diag\left(\sigma_1^2, \sigma_2^2, \ldots, \sigma_n^2\right)$ and denote its true value by $\boldsymbol{\Sigma}_0 = Diag\left(\sigma_{10}^2, \sigma_{20}^2, \ldots, \sigma_{n0}^2\right)$. Eq. (21) is a valid moment condition under unknown forms of heteroskedasticity since at the true value $\boldsymbol{\psi}_0$ we have

$$E_0\left[\boldsymbol{\varepsilon}'\left(\boldsymbol{\psi}_0\right)\mathbf{C}\boldsymbol{\varepsilon}\left(\boldsymbol{\psi}_0\right)\right] = E_0\left(\boldsymbol{\varepsilon}'\mathbf{C}\boldsymbol{\varepsilon}\right) = Tr\left(\mathbf{C}\boldsymbol{\Sigma}_0\right) = Tr\left(\mathbf{B}\boldsymbol{\Sigma}_0\right) = \sum_{i=1}^{n} b_{ii}\sigma_{i0}^2 = 0,$$

which does not require $\boldsymbol{\Sigma}_0$ to be known. Here we consider a single quadratic moment for ease of exposition. In practice, one could use multiple quadratic moment conditions, namely $E_0\left(\boldsymbol{\varepsilon}'\mathbf{C}_\ell\boldsymbol{\varepsilon}\right) = 0$, for $\ell = 1, 2, \ldots, L$, where L is a finite number, $\mathbf{C}_\ell = \left(\mathbf{B}_\ell + \mathbf{B}_\ell'\right)/2$, and $\mathbf{B}_\ell$ satisfies the conditions of Assumption 7.

In the special case where the errors are homoskedastic, Assumption 7(a) can be replaced by the weaker requirement, $Tr\left(\mathbf{B}\right) = 0$. As pointed out by Lin and Lee (2010), using a $\mathbf{B}$ matrix with zero diagonal elements will produce consistent estimates irrespective of whether the errors are homoskedastic or heteroskedastic, but it will be asymptotically less efficient if the errors happen to be homoskedastic. On the other hand, under heteroskedasticity using a $\mathbf{B}$ matrix with zero trace but non-zero diagonal entries will yield inconsistent estimates.

Lee (2007) assumes that $\mathbf{B}$ is uniformly bounded in both row and column sums in absolute value and suggests using $\mathbf{B}_\ell = \mathbf{W}^\ell - n^{-1}Tr\left(\mathbf{W}^\ell\right)\mathbf{I}_n$, for $\ell = 1, 2, \ldots, L$, in the quadratic moments if the errors are homoskedastic, where $\mathbf{W}^\ell$ denotes the ℓth power of $\mathbf{W}$. If the errors are heteroskedastic, practical choices of $\mathbf{B}_\ell$ can be $\mathbf{B}_\ell = \mathbf{W}^\ell - Diag\left(\mathbf{W}^\ell\right)$. In contrast, in our set up where columns of $\mathbf{W}$ need not be bounded (see Assumption 4), in part (c) of Assumption 7 we have relaxed Lee's boundedness condition on $\mathbf{B}$, and allow the column norm of $\mathbf{B}$ to rise with n at the rate of δ_b.

We are now ready to define the GMM estimator of $\boldsymbol{\psi}_0$ of model (1), denoted by $\tilde{\boldsymbol{\psi}} = (\tilde{\rho}, \tilde{\boldsymbol{\beta}}')'$, using both quadratic and linear moment conditions:

$$\tilde{\boldsymbol{\psi}} = \arg\min_{\boldsymbol{\psi}\in\boldsymbol{\Psi}} \mathbf{g}_n'(\boldsymbol{\psi})\left(\mathbf{A}_n'\mathbf{A}_n\right)\mathbf{g}_n\left(\boldsymbol{\psi}\right), \tag{23}$$

where $\mathbf{g}_n(\boldsymbol{\psi})$ is a $(k+1)\times 1$ vector given by

$$\mathbf{g}_n(\boldsymbol{\psi}) = \begin{pmatrix} n^{-1}\boldsymbol{\varepsilon}'\left(\boldsymbol{\psi}\right)\mathbf{C}\boldsymbol{\varepsilon}\left(\boldsymbol{\psi}\right) \\ n^{-1}\mathbf{Z}'\boldsymbol{\varepsilon}\left(\boldsymbol{\psi}\right) \end{pmatrix}, \tag{24}$$

and $\mathbf{A}_n$ is a $(k+1)\times(r+1)$ matrix of full row rank, assumed to converge to a constant full row rank matrix $\mathbf{A}$.

Before proceeding to examine the asymptotic properties of $\tilde{\boldsymbol{\psi}}$, we first focus on the problem of identification in the case of pure SAR models without exogenous regressors. In this case (1) simplifies to,

$$\mathbf{y} = \rho\mathbf{y}^* + \boldsymbol{\varepsilon}, \tag{25}$$

with Assumption 2 replaced by

Assumption 8. The parameter ρ of model (25) satisfies $\rho \in \boldsymbol{\Theta}_\rho$, where $\boldsymbol{\Theta}_\rho$ is a compact subset of $\mathbb{R}$. The true value of ρ, denoted by ρ_0, lies in the interior of the parameter space, $\boldsymbol{\Theta}_\rho$.

The GMM estimator of ρ_0 in model (25) can be obtained by

$$\tilde{\rho} = \arg\min_{\rho\in\boldsymbol{\Theta}_\rho} g_n^2(\rho), \tag{26}$$

where $g_n(\rho) = n^{-1}\boldsymbol{\varepsilon}'(\rho)\mathbf{C}\boldsymbol{\varepsilon}(\rho)$, and $\boldsymbol{\varepsilon}(\rho) = \mathbf{y} - \rho\mathbf{y}^*$. Proposition 1 shows that in order to uniquely identify ρ_0 in the pure SAR model (25), at least two moment conditions are required. Specifically, the GMM estimator of ρ_0 based on L quadratic moments (L is a finite number) is given by

$$\tilde{\rho} = \arg\min_{\rho\in\boldsymbol{\Theta}_\rho} \left[\mathbf{a}_n'\mathbf{g}_n(\rho)\right]^2, \tag{27}$$

where $\mathbf{g}_n(\rho) = \left[g_{1,n}(\rho), g_{2,n}(\rho), \ldots, g_{L,n}(\rho)\right]'$, $g_{\ell,n}(\rho) = n^{-1}\boldsymbol{\varepsilon}'(\rho)\mathbf{C}_\ell\boldsymbol{\varepsilon}(\rho)$, for $\ell = 1, 2, \ldots, L$, and $\mathbf{a}_n$ is an $L \times 1$ non-zero and non-negative vector.

Proposition 1. *Consider the SAR model given by (25), and suppose that Assumptions 1, 4, 7 (a)–(c), and 8 hold. Then to uniquely identify ρ_0 it is required that the GMM estimator, defined by (27), is based on at least two independent quadratic moment conditions, in the sense that the ratios $b_{\ell 0}/a_{\ell 0}$, are not all the same across $\ell = 1, 2, \ldots, L \geq 2$, where $a_{\ell 0} = \lim_{n\to\infty} Tr\left(n^{-1}\mathbf{G}_0'\mathbf{C}_\ell\mathbf{G}_0\boldsymbol{\Sigma}_0\right)$, and $b_{\ell 0} = \lim_{n\to\infty} Tr\left(n^{-1}\mathbf{G}_0'\mathbf{C}_\ell\boldsymbol{\Sigma}_0\right)$.*

See Section S1.2 of the Online Supplement for a proof.

Remark 6. When the GMM estimator is based on a single quadratic moment condition, the parameter ρ_0 of model (25) is not uniquely identified and the GMM estimator of ρ computed by minimizing $g_n^2(\rho)$ defined by (26), converges in probability to ρ_0 or $\rho_0 + 2b_0/a_0$, where $a_0 = \lim_{n\to\infty} Tr\left(n^{-1}\mathbf{G}_0'\mathbf{C}\mathbf{G}_0\boldsymbol{\Sigma}_0\right)$ and $b_0 = \lim_{n\to\infty} Tr\left(n^{-1}\mathbf{G}_0'\mathbf{C}\boldsymbol{\Sigma}_0\right)$. In practice, it is advisable to use at least two quadratic moments if the SAR model does not contain any exogenous regressors.

Consider now the SAR model given by (1) that includes exogenous regressors. For ease of exposition, in what follows we set $\delta_b = \delta$, that is, $\|\mathbf{B}\|_1$ rises with n at a rate equal to or slower than the rate of $\|\mathbf{W}\|_1$, since in practice $\mathbf{W}$ is commonly adopted as the $\mathbf{B}$ matrix. The following theorem shows that $\boldsymbol{\psi}_0 = (\rho_0, \boldsymbol{\beta}_0')$ of model (1) can be globally identified if we have enough instruments such that the rank condition in Assumption 6(c) holds. The theorem also establishes consistency and asymptotic normality of the GMM estimator defined by (23).

Theorem 3. *Consider the SAR model given by (1). Suppose that Assumptions 1–4, 6 and 7 hold, with $\delta_b = \delta$. Then*

(a) $\boldsymbol{\psi}_0 = (\rho_0, \boldsymbol{\beta}_0')$ is globally identified,

(b) the GMM estimator of $\boldsymbol{\psi}_0$, denoted by $\tilde{\boldsymbol{\psi}}$ and defined in (23), is consistent for $\boldsymbol{\psi}_0$ if δ (the degree of network centrality) defined by (4) lies in the range $0 \leq \delta < 1$,

(c) $\sqrt{n}\left(\tilde{\boldsymbol{\psi}} - \boldsymbol{\psi}_0\right)$ is asymptotically normally distributed as $n \to \infty$, if δ lies in the range $0 \leq \delta < 1/2$, namely,

$$\sqrt{n}\left(\tilde{\boldsymbol{\psi}} - \boldsymbol{\psi}_0\right) \to_d N\left[\mathbf{0},\ \left(\mathbf{D}'\boldsymbol{S}_{aa}\mathbf{D}\right)^{-1}\left(\mathbf{D}'\boldsymbol{S}_{aa}\mathbf{V}_g\boldsymbol{S}_{aa}\mathbf{D}\right)\left(\mathbf{D}'\boldsymbol{S}_{aa}\mathbf{D}\right)^{-1}\right],$$

where $\boldsymbol{S}_{aa} = \lim_{n\to\infty}\mathbf{A}_n'\mathbf{A}_n$, $\mathbf{A}_n$ is the $(k+1)\times(r+1)$ matrix of full row rank used in the GMM minimand, (23),

$$\mathbf{D} = \left\{\left[2\lim_{n\to\infty} Tr\left(n^{-1}\boldsymbol{\Sigma}_0\mathbf{C}\mathbf{G}_0\right), \mathbf{0}_{1\times k}\right]', \boldsymbol{\Sigma}_{zq_0}'\right\}', \tag{28}$$

$$\mathbf{V}_g = \begin{pmatrix} 2\lim_{n\to\infty} Tr\left[n^{-1}\left(\boldsymbol{\Sigma}_0\mathbf{C}\right)^2\right] & \mathbf{0}_{1\times r} \\ \mathbf{0}_{r\times 1} & p\lim_{n\to\infty} n^{-1}\mathbf{Z}'\boldsymbol{\Sigma}_0\mathbf{Z} \end{pmatrix}, \tag{29}$$

$\mathbf{G}_0$ is defined by (8), $\boldsymbol{\Sigma}_{zq_0} = p\lim_{n\to\infty} n^{-1}\mathbf{Z}'\mathbf{Q}_0$, and $\boldsymbol{\Sigma}_0 = Diag\left(\sigma_{10}^2, \sigma_{20}^2, \ldots, \sigma_{n0}^2\right)$.

The proof of Theorem 3 is given in Section S1.2 of the Online Supplement.

A few comments on Theorem 3 are in order. First, it is worth emphasizing that $\boldsymbol{\psi}_0$ is globally identified if we have enough instruments such that the rank condition in Assumption 6 (c) is satisfied, irrespective of the number of quadratic moments included. Using a finite number of quadratic moments in addition to linear moments improves efficiency.

Second, Theorem 3 is consistent with the result in Lin and Lee (2010), who extended the GMM estimator for SAR models proposed by Lee (2007) to the case of unknown error heteroskedasticity assuming $\delta = 0$.

Third, if the errors are identically distributed with variances $\sigma_{i0}^2 = \sigma_0^2$, for all i, and Assumption 7(a) is replaced by $Tr(\mathbf{B}) = 0$, then the main conclusions in Theorem 3 are unaffected except that the expressions in (28) and (29) simplify as follows:

$$\mathbf{D} = \left[\left(2\sigma_0^2 \lim_{n\to\infty} Tr\left(n^{-1}\mathbf{G}_0'\mathbf{C}\right), \mathbf{0}_{1\times k}\right)', \boldsymbol{\Sigma}_{zq_0}'\right]', \quad \mathbf{V}_g = \begin{pmatrix} v_{10} & \mu_{30}\boldsymbol{\upsilon}' \\ \mu_{30}\boldsymbol{\upsilon} & \sigma_0^2\boldsymbol{\Sigma}_{zz} \end{pmatrix}, \tag{30}$$

$$v_{10} = \lim_{n\to\infty}\left[\gamma_{20}n^{-1}\sum_{i=1}^{n} c_{ii}^2 + 2\sigma_0^4 Tr\left(n^{-1}\mathbf{C}^2\right)\right],$$

where $\boldsymbol{\Sigma}_{zz} = p\lim_{n\to\infty} n^{-1}\mathbf{Z}'\mathbf{Z}$, $\boldsymbol{\upsilon} = p\lim_{n\to\infty} n^{-1}\mathbf{Z}'[diag(\mathbf{C})]$, $\mu_{30} = E(\varepsilon_i^3)$, $\gamma_{20} = E(\varepsilon_i^4) - 3\sigma_0^4$, and c_{ii} is the ith diagonal element of $\mathbf{C}$. Observe that the asymptotic distribution given by Theorem 3(c) does not depend on the third and fourth order moments of the errors since $\mathbf{B}$ is restricted so that $diag(\mathbf{B}) = \mathbf{0}$. In contrast, (30) which assumes homoskedastic errors and only requires $Tr(\mathbf{B}) = 0$, does involve higher-order moments of the errors, ε_i.

Remark 7. To consistently estimate the variance of the GMM estimator, one can replace $\boldsymbol{\Sigma}$ with $\hat{\boldsymbol{\Sigma}} = Diag\left(\tilde{\varepsilon}_1^2, \tilde{\varepsilon}_2^2, \ldots, \tilde{\varepsilon}_n^2\right)$, where $\tilde{\varepsilon}_i = y_i - \tilde{\rho}y_i^* - \tilde{\boldsymbol{\beta}}'\mathbf{x}_i$. In the case of homoskedastic errors and if $diag(\mathbf{B}) \neq \mathbf{0}$, consistent estimators of μ_{30} and γ_{20} are given by $\tilde{\mu}_3 = n^{-1}\sum_{i=1}^n \tilde{\varepsilon}_i^3$ and $\tilde{\gamma}_2 = n^{-1}\sum_{i=1}^n \tilde{\varepsilon}_i^4 - 3\left(\tilde{\sigma}^2\right)^2$, respectively, where $\tilde{\sigma}^2 = n^{-1}\sum_{j=1}^n \tilde{\varepsilon}_j^2$.

As is well known in the GMM literature, the optimal moments weighting matrix is given by $\mathbf{V}_g^{-1}$. A feasible optimal GMM (OGMM) estimator of $\boldsymbol{\psi}_0$, denoted by $\tilde{\boldsymbol{\psi}}_{opt}$, can be obtained by using a consistent estimator of $\mathbf{V}_g^{-1}$ for $\mathbf{A}_n'\mathbf{A}_n$, that is,

$$\tilde{\boldsymbol{\psi}}_{opt} = \arg\min_{\boldsymbol{\psi}\in\boldsymbol{\Psi}} \mathbf{g}_n'(\boldsymbol{\psi})\tilde{\mathbf{V}}_g^{-1}\mathbf{g}_n'(\boldsymbol{\psi}), \tag{31}$$

where $\mathbf{g}_n(\boldsymbol{\psi})$ is given by (24) and $\tilde{\mathbf{V}}_g$ is a consistent estimator of $\mathbf{V}_g$. Then $\tilde{\boldsymbol{\psi}}_{opt}$ is consistent for $\boldsymbol{\psi}_0$ when δ is in the range $0 \leq \delta < 1$, and it has the following asymptotic distribution as $n \to \infty$ when δ lies in the range $0 \leq \delta < 1/2$,

$$\sqrt{n}\left(\tilde{\boldsymbol{\psi}}_{opt} - \boldsymbol{\psi}_0\right) \to_d N\left[\mathbf{0}, \left(\mathbf{D}'\mathbf{V}_g^{-1}\mathbf{D}\right)^{-1}\right].$$

The best choice of $\mathbf{B}$ exists under certain conditions. Lee (2007) shows that if the idiosyncratic errors are identically and normally distributed, the OGMM estimator using $\mathbf{G}_0 - n^{-1}Tr(\mathbf{G}_0)\mathbf{I}_n$ in the quadratic moment condition and $\mathbf{Q}_0 = \left(\mathbf{G}_0\mathbf{X}\boldsymbol{\beta}_0, \mathbf{X}\right)$ in the linear moment conditions, has the smallest asymptotic variance among the set of GMM estimators derived with the class of matrices, $\mathbf{B}_\ell$, having zero trace. Among the group of GMM estimators derived with the class of matrices having zero diagonal elements, the OGMM estimator using $\mathbf{G}_0 - Diag(\mathbf{G}_0)$ and $\mathbf{Q}_0$ in the moments has the smallest asymptotic variance.[9] The OGMM estimator using $\mathbf{G}_0 - n^{-1}Tr(\mathbf{G}_0)\mathbf{I}_n$ (or $\mathbf{G}_0 - Diag(\mathbf{G}_0)$) and $\mathbf{Q}_0$ is referred to as the best GMM estimator. By a similar argument and applying Lemma S.6 of the Online Supplement, it is straightforward to show that the asymptotic properties of the best GMM estimator can be extended to the case where the column sums of $\mathbf{W}$ rise with n, under the same conditions on δ as in Theorem 3. Since both $\mathbf{G}_0$ and $\mathbf{Q}_0$ depend on unknown parameters, a feasible best GMM estimator can be implemented in two steps: In the first step, a preliminary consistent estimate of $\boldsymbol{\psi}_0$ is obtained, which is then used in the second step to compute the optimal GMM estimates using $\mathbf{Q}_0$ and $\mathbf{G}_0 - n^{-1}Tr(\mathbf{G}_0)\mathbf{I}_n$ if assuming homoskedasticity, or $\mathbf{G}_0 - Diag(\mathbf{G}_0)$ if assuming heteroskedasticity, evaluated at the first-stage estimates. In the rest of this paper we focus on the feasible best GMM estimator and refer to it simply as the GMM estimator, for brevity.[10]

5. BMM estimation

In this section we develop the bias-corrected method of moments (BMM) estimator of $\boldsymbol{\psi}_0 = \left(\rho_0, \boldsymbol{\beta}_0'\right)'$ for the SAR model given by (1). The BMM procedure uses least squares but corrects the bias due to the endogeneity of the spatial variable, $\mathbf{y}^*$. To clarify the idea, let us first consider the case of homoskedastic errors and then relax it to allow for unknown heteroskedasticity. Let $\boldsymbol{\theta}_0 = (\boldsymbol{\psi}_0', \sigma_0^2)' = (\rho_0, \boldsymbol{\beta}_0', \sigma_0^2)'$ denote the vector of unknown parameters of the SAR model under homoskedasticity, where $\varepsilon_i \sim IID\left(0, \sigma_0^2\right)$ and $\sup_i E|\varepsilon_i|^{4+c} < K$, for some $c > 0$. The application of BMM to the SAR model is straightforward. Using $\mathbf{y}^*$ and $\mathbf{X}$ as instruments, the bias-corrected population moments are given by

$$E\left[\mathbf{y}^{*\prime}\left(\mathbf{y} - \rho\mathbf{y}^* - \mathbf{X}\boldsymbol{\beta}\right)\right] = E\left(\mathbf{y}^{*\prime}\boldsymbol{\varepsilon}\right), \tag{32}$$

$$E\left[\mathbf{X}'\left(\mathbf{y} - \rho\mathbf{y}^* - \mathbf{X}\boldsymbol{\beta}\right)\right] = \mathbf{0}, \tag{33}$$

$$E\left[\left(\mathbf{y} - \rho\mathbf{y}^* - \mathbf{X}\boldsymbol{\beta}\right)'\left(\mathbf{y} - \rho\mathbf{y}^* - \mathbf{X}\boldsymbol{\beta}\right)\right] = n\sigma^2. \tag{34}$$

[9] This result does not require the condition that the idiosyncratic errors are normally distributed. See Lee (2007) Proposition 3 for details.

[10] We also examined the finite sample properties of GMM estimators that use other instruments and $\mathbf{B}$ matrices. The results are provided in the Online Supplement, and as can be seen overall they have less satisfactory small sample properties.

Using (7), we have $E\left(\mathbf{y}^{*\prime}\boldsymbol{\varepsilon}\right)=E\left[\left(\boldsymbol{\beta}^{\prime}\mathbf{X}^{\prime}+\boldsymbol{\varepsilon}^{\prime}\right)\mathbf{G}^{\prime}\left(\rho\right)\boldsymbol{\varepsilon}\right]$, and under Assumption 3(a) and the IID assumption of ε_i, we obtain $E\left(\mathbf{y}^{*\prime}\boldsymbol{\varepsilon}\right)=\sigma^2 Tr\left[\mathbf{G}\left(\rho\right)\right]$. The sample version of the moment conditions (32)–(34) can now be written as

$$n^{-1}\mathbf{y}^{*\prime}\left(\mathbf{y}-\hat{\rho}\mathbf{y}^{*}-\mathbf{X}\hat{\boldsymbol{\beta}}\right)=\hat{\sigma}^{2}Tr\left[n^{-1}\mathbf{G}\left(\hat{\rho}\right)\right], \tag{35}$$

$$n^{-1}\mathbf{X}^{\prime}\left(\mathbf{y}-\hat{\rho}\mathbf{y}^{*}-\mathbf{X}\hat{\boldsymbol{\beta}}\right)=\mathbf{0}, \tag{36}$$

$$n^{-1}\left(\mathbf{y}-\hat{\rho}\mathbf{y}^{*}-\mathbf{X}\hat{\boldsymbol{\beta}}\right)^{\prime}\left(\mathbf{y}-\hat{\rho}\mathbf{y}^{*}-\mathbf{X}\hat{\boldsymbol{\beta}}\right)=\hat{\sigma}^{2}. \tag{37}$$

Let $\hat{\boldsymbol{\theta}}=(\hat{\rho},\hat{\boldsymbol{\beta}}^{\prime},\hat{\sigma}^{2})^{\prime}$ denote the BMM estimator of $\boldsymbol{\theta}_0$. The system of Eqs. (35)–(37) can now be used to solve for $\hat{\boldsymbol{\theta}}$ as follows:

$$\hat{\boldsymbol{\theta}}=\arg\min_{\boldsymbol{\theta}\in\boldsymbol{\Theta}}\mathbf{m}_{n}^{\prime}\left(\boldsymbol{\theta}\right)\mathbf{m}_{n}\left(\boldsymbol{\theta}\right), \tag{38}$$

where $\mathbf{m}_{n}\left(\boldsymbol{\theta}\right)=\left[m_{1,n}\left(\boldsymbol{\theta}\right),\mathbf{m}_{2,n}^{\prime}\left(\boldsymbol{\theta}\right),m_{3,n}\left(\boldsymbol{\theta}\right)\right]^{\prime}$ with $m_{1,n}\left(\boldsymbol{\theta}\right)=n^{-1}\mathbf{y}^{*\prime}\boldsymbol{\varepsilon}\left(\boldsymbol{\psi}\right)-\sigma^{2}Tr\left[n^{-1}\mathbf{G}\left(\rho\right)\right]$, $\mathbf{m}_{2,n}\left(\boldsymbol{\theta}\right)=n^{-1}\mathbf{X}^{\prime}\boldsymbol{\varepsilon}\left(\boldsymbol{\psi}\right)$, $m_{3,n}\left(\boldsymbol{\theta}\right)=n^{-1}\boldsymbol{\varepsilon}^{\prime}\left(\boldsymbol{\psi}\right)\boldsymbol{\varepsilon}\left(\boldsymbol{\psi}\right)-\sigma^{2}$, $\boldsymbol{\varepsilon}\left(\boldsymbol{\psi}\right)$ is given by (19), $\boldsymbol{\Theta}=\boldsymbol{\Theta}_{\rho}\times\boldsymbol{\Theta}_{\beta}\times\boldsymbol{\Theta}_{\sigma^2}$, and $\boldsymbol{\Theta}_{\sigma^2}$ is a compact subspace of $(0,\infty)$ containing the true value σ_0^2.

Unlike least squares, the BMM procedure is non-linear in $\hat{\rho}$, and its asymptotic properties depend on the assumptions regarding the rate at which the column sums of $\mathbf{W}$ rise with n. As we shall see, the BMM estimators are consistent and do not suffer from the weak instrument problem since $\mathbf{y}^{*}$ is instrumented with its own values. However, in small samples it might be beneficial to augment the system of estimating equations, (35)–(37), with additional moment conditions.

We show in the Online Supplement that the BMM estimator under homoskedasticity, defined by (38), is a consistent estimator when δ is in the range $0\leq\delta<1$, where as before δ is a measure of network centrality, defined by (4). Moreover, when δ is in the range $0\leq\delta<1/2$, $\sqrt{n}\left(\hat{\boldsymbol{\psi}}-\boldsymbol{\psi}_0\right)$ is asymptotically normally distributed:

$$\sqrt{n}\left(\hat{\boldsymbol{\psi}}-\boldsymbol{\psi}_0\right)\rightarrow_{d}N\left(\mathbf{0},\mathbf{H}^{-1}\mathbf{V}\mathbf{H}^{-1}\right), \tag{39}$$

as $n\rightarrow\infty$, where

$$\mathbf{H}=\begin{pmatrix}\boldsymbol{\beta}_0^{\prime}\boldsymbol{\Sigma}_{xg_0g_0x}\boldsymbol{\beta}_0+\sigma_0^2h_0 & \boldsymbol{\beta}_0^{\prime}\boldsymbol{\Sigma}_{xg_0x}\\ \boldsymbol{\Sigma}_{xg_0x}\boldsymbol{\beta}_0 & \boldsymbol{\Sigma}_{xx}\end{pmatrix},\quad \mathbf{V}=\begin{pmatrix}q_0^2 & \sigma_0^2\boldsymbol{\beta}_0^{\prime}\boldsymbol{\Sigma}_{xg_0x}\\ \sigma_0^2\boldsymbol{\Sigma}_{xg_0x}\boldsymbol{\beta}_0 & \sigma_0^2\boldsymbol{\Sigma}_{xx}\end{pmatrix}, \tag{40}$$

$$q_0^2=\sigma_0^2\boldsymbol{\beta}_0^{\prime}\boldsymbol{\Sigma}_{xg_0g_0x}\boldsymbol{\beta}_0+\gamma_{20}\,p\lim_{n\rightarrow\infty}n^{-1}\sum_{i=1}^{n}\pi_{ii,0}^2+2\mu_{30}\,p\lim_{n\rightarrow\infty}n^{-1}\left[diag\left(\boldsymbol{\Pi}_0\right)\right]^{\prime}\mathbf{G}_0\mathbf{X}\boldsymbol{\beta}_0 \tag{41}$$

$$+\sigma_0^4 p\lim_{n\rightarrow\infty}\left[Tr\left(n^{-1}\boldsymbol{\Pi}_0^{\prime}\boldsymbol{\Pi}_0\right)+Tr\left(n^{-1}\boldsymbol{\Pi}_0^2\right)\right],$$

$$\boldsymbol{\Pi}_0=\mathbf{G}_0-\mathbf{M}_xTr\left(n^{-1}\mathbf{G}_0\right),\ \ \mathbf{M}_x=\mathbf{I}_n-\mathbf{X}\left(\mathbf{X}^{\prime}\mathbf{X}\right)^{-1}\mathbf{X}^{\prime}, \tag{42}$$

$$h_0=\lim_{n\rightarrow\infty}\left\{n^{-1}Tr\left(\mathbf{G}_0^2+\mathbf{G}_0^{\prime}\mathbf{G}_0\right)-2n^{-2}\left[Tr\left(\mathbf{G}_0\right)\right]^2\right\}, \tag{43}$$

where $\mathbf{G}_0$ is defined by (8), $\pi_{ii,0}$ is the ith diagonal element of $\boldsymbol{\Pi}_0$, $\mu_{30}=E(\varepsilon_i^3)$, and $\gamma_{20}=E(\varepsilon_i^4)-3\sigma_0^4$.

Remark 8. It can be seen from (41) that the variance formula will not involve the third and fourth moments of the error term if (i) ε_i is Gaussian, since under Gaussianity $\gamma_{20}=0$ and $\mu_{30}=0$; or (ii) the diagonal elements of $\boldsymbol{\Pi}_0$ are zero, which occurs if $\mathbf{G}_0$ has zero diagonal entries. Furthermore, the variances of both BMM and GMM estimators under homoskedasticity will not involve the third moment of the error term if the model does not contain $\mathbf{X}$. In general, μ_{30} can be estimated by $\hat{\mu}_3=n^{-1}\sum_{i=1}^{n}\hat{\varepsilon}_i^3$, where $\hat{\varepsilon}_i=y_i-\hat{\rho}y_i^{*}-\hat{\boldsymbol{\beta}}^{\prime}\mathbf{x}_i$, and γ_{20} can be estimated by $\hat{\gamma}_2=n^{-1}\sum_{i=1}^{n}\hat{\varepsilon}_i^4-3\left(\hat{\sigma}^2\right)^2$.

We now turn to considering the BMM estimation under general forms of heteroskedasticity where the errors satisfy Assumption 1. Recall that $\boldsymbol{\Sigma}=Diag\left(\sigma_1^2,\sigma_2^2,\ldots,\sigma_n^2\right)$ and note that

$$E\left(\mathbf{y}^{*\prime}\boldsymbol{\varepsilon}\right)=E\left[\left(\boldsymbol{\beta}^{\prime}\mathbf{X}^{\prime}+\boldsymbol{\varepsilon}^{\prime}\right)\mathbf{G}^{\prime}\left(\rho\right)\boldsymbol{\varepsilon}\right]=Tr\left[\mathbf{G}\left(\rho\right)\boldsymbol{\Sigma}\right]=\sum_{i=1}^{n}g_{ii}\left(\rho\right)\sigma_i^2, \tag{44}$$

where $g_{ii}\left(\rho\right)$ is the ith diagonal element of $\mathbf{G}\left(\rho\right)$. In view of (44), the moment condition (32) becomes $E\left[\mathbf{y}^{*\prime}\left(\mathbf{y}-\rho\mathbf{y}^{*}-\mathbf{X}\boldsymbol{\beta}\right)\right]=\sum_{i=1}^{n}g_{ii}\left(\rho\right)\sigma_i^2$, which can be written equivalently as

$$E\left[\mathbf{y}^{*\prime}\left(\mathbf{y}-\rho\mathbf{y}^{*}-\mathbf{X}\boldsymbol{\beta}\right)\right]=E\left[\sum_{i=1}^{n}g_{ii}\left(\rho\right)\varepsilon_i^2\right]=E\left[\boldsymbol{\varepsilon}^{\prime}\check{\mathbf{G}}\left(\rho\right)\boldsymbol{\varepsilon}\right], \tag{45}$$

where $\check{\mathbf{G}}(\rho) = Diag[\mathbf{G}(\rho)]$.[11] In sum, the BMM population moment conditions under heteroskedasticity are given by (33) and (45), and their sample counterparts are

$$n^{-1}\mathbf{y}^{*\prime}\left(\mathbf{y}-\hat{\rho}\mathbf{y}^{*}-\mathbf{X}\hat{\boldsymbol{\beta}}\right)=n^{-1}\left(\mathbf{y}-\hat{\rho}\mathbf{y}^{*}-\mathbf{X}\hat{\boldsymbol{\beta}}\right)^{\prime}\check{\mathbf{G}}(\hat{\rho})\left(\mathbf{y}-\hat{\rho}\mathbf{y}^{*}-\mathbf{X}\hat{\boldsymbol{\beta}}\right), \tag{46}$$

$$n^{-1}\mathbf{X}^{\prime}\left(\mathbf{y}-\hat{\rho}\mathbf{y}^{*}-\mathbf{X}\hat{\boldsymbol{\beta}}\right)=\mathbf{0}. \tag{47}$$

Now let $\hat{\boldsymbol{\psi}}=\left(\hat{\rho},\hat{\boldsymbol{\beta}}^{\prime}\right)^{\prime}$ denote the BMM estimator of $\boldsymbol{\psi}_0$ assuming heteroskedastic errors. To avoid introducing further notations, we use the same notations as in the case of homoskedastic errors. The BMM estimator of $\boldsymbol{\psi}_0$, is obtained by solving the system of Eqs. (46) and (47) for $\hat{\boldsymbol{\psi}}$, which can be written equivalently as

$$\hat{\boldsymbol{\psi}}=\arg\min_{\boldsymbol{\psi}\in\boldsymbol{\Psi}}\mathbf{m}_n^{\prime}(\boldsymbol{\psi})\,\mathbf{m}_n(\boldsymbol{\psi}), \tag{48}$$

where $\mathbf{m}_n(\boldsymbol{\psi})=\left[m_{1,n}(\boldsymbol{\psi}),\mathbf{m}_{2,n}^{\prime}(\boldsymbol{\psi})\right]^{\prime}$ with $m_{1,n}(\boldsymbol{\psi})=n^{-1}\mathbf{y}^{*\prime}\boldsymbol{\varepsilon}(\boldsymbol{\psi})-n^{-1}\boldsymbol{\varepsilon}^{\prime}(\boldsymbol{\psi})\check{\mathbf{G}}(\rho)\boldsymbol{\varepsilon}(\boldsymbol{\psi})$, $\mathbf{m}_{2,n}(\boldsymbol{\psi})=n^{-1}\mathbf{X}^{\prime}\boldsymbol{\varepsilon}(\boldsymbol{\psi})$, and as before $\boldsymbol{\varepsilon}(\boldsymbol{\psi})$ is given by (19).

The following theorem formally summarizes the asymptotic distribution of the BMM estimator assuming heteroskedasticity. Its proof is given in Section S1.2 of the Online Supplement.

Theorem 4. *Consider the SAR model given by (1), and suppose that Assumptions 1–5 hold. Then*

(a) the bias-corrected method of moments (BMM) estimator of $\boldsymbol{\psi}_0=\left(\rho_0,\boldsymbol{\beta}_0^{\prime}\right)^{\prime}$, *denoted by* $\hat{\boldsymbol{\psi}}=(\hat{\rho},\hat{\boldsymbol{\beta}}^{\prime})^{\prime}$ *and defined by (48), is consistent for* $\boldsymbol{\psi}_0$ *when* δ, *the centrality of the weights matrix* $\mathbf{W}$ *defined by (4), lies in the range* $0\leq\delta<1$.

(b) $\sqrt{n}\left(\hat{\boldsymbol{\psi}}-\boldsymbol{\psi}_0\right)$ *is asymptotically normally distributed as* $n\to\infty$, *when* δ *is in the range* $0\leq\delta<1/2$, *namely*

$$\sqrt{n}\left(\hat{\boldsymbol{\psi}}-\boldsymbol{\psi}_0\right)\to_d N\left(\mathbf{0},\mathbf{H}^{-1}\mathbf{V}\mathbf{H}^{-1}\right), \tag{49}$$

where

$$\mathbf{H}=\begin{pmatrix}\boldsymbol{\beta}_0^{\prime}\boldsymbol{\Sigma}_{xg_0g_0x}\boldsymbol{\beta}_0+h_0 & \boldsymbol{\beta}_0^{\prime}\boldsymbol{\Sigma}_{xg_0x}\\ \boldsymbol{\Sigma}_{xg_0x}\boldsymbol{\beta}_0 & \boldsymbol{\Sigma}_{xx}\end{pmatrix},\quad \mathbf{V}=p\lim_{n\to\infty}\begin{pmatrix}q_0^2 & n^{-1}\boldsymbol{\beta}_0^{\prime}\mathbf{X}^{\prime}\mathbf{G}_0^{\prime}\boldsymbol{\Sigma}_0\mathbf{X}\\ n^{-1}\mathbf{X}^{\prime}\boldsymbol{\Sigma}_0\mathbf{G}_0\mathbf{X}\boldsymbol{\beta}_0 & n^{-1}\mathbf{X}^{\prime}\boldsymbol{\Sigma}_0\mathbf{X}\end{pmatrix}, \tag{50}$$

$$q_0^2=n^{-1}\boldsymbol{\beta}_0^{\prime}\mathbf{X}^{\prime}\mathbf{G}_0^{\prime}\boldsymbol{\Sigma}_0\mathbf{G}_0\mathbf{X}\boldsymbol{\beta}_0+Tr\left[n^{-1}\boldsymbol{\Sigma}_0\boldsymbol{\Pi}_0\left(\boldsymbol{\Sigma}_0\boldsymbol{\Pi}_0+\boldsymbol{\Sigma}_0\boldsymbol{\Pi}_0^{\prime}\right)\right],\quad \boldsymbol{\Pi}_0=\mathbf{G}_0-\check{\mathbf{G}}_0, \tag{51}$$

$$h_0=\lim_{n\to\infty}Tr\left[n^{-1}\left(\mathbf{G}_0^2+\mathbf{G}_0^{\prime}\mathbf{G}_0-2\check{\mathbf{G}}_0\mathbf{G}_0\right)\boldsymbol{\Sigma}_0\right], \tag{52}$$

$\mathbf{G}_0$ *is given by (8),* $\check{\mathbf{G}}_0=Diag(\mathbf{G}_0)$, *and* $\boldsymbol{\Sigma}_0=Diag\left(\sigma_{10}^2,\sigma_{20}^2,\ldots,\sigma_{n0}^2\right)$.

A number of remarks are in order. First, it should be noted that unlike linear regressions, but similar to the GMM estimator discussed above, the BMM estimator assuming homoskedasticity given by (48) will result in inconsistent estimates if the errors are in fact heteroskedastic. On the other hand, if the errors are homoskedastic, the BMM estimator assuming heteroskedasticity given by (38) will produce consistent but asymptotically less efficient estimates.

Second, it is clear from (49) that $\boldsymbol{\psi}_0$ is identified if $\mathbf{H}$, defined in (50), is positive definite. Notice that $\mathbf{H}=\mathbf{H}_1+\mathbf{H}_2$, where $\mathbf{H}_1=p\lim_{n\to\infty}n^{-1}\mathbf{Q}_0^{\prime}\mathbf{Q}_0$, and

$$\mathbf{H}_2=\begin{pmatrix}h_0 & \mathbf{0}_{1\times k}\\ \mathbf{0}_{k\times 1} & \mathbf{0}_{k\times k}\end{pmatrix}.$$

Since $\mathbf{H}_1$ is positive semi-definite and $h_0\geq 0$, it follows that $\mathbf{H}$ is positive definite if either $h_0>0$ and/or if $\mathbf{H}_1$ is positive definite. Therefore, Assumption 5 ensures that $\boldsymbol{\psi}_0$ is identified. Moreover, if $\rho_0\geq 0$, the identification condition given by Assumption 5(b) is ensured if

$$Tr\left[n^{-1}\left(\mathbf{G}_0^2+\mathbf{G}_0^{\prime}\mathbf{G}_0-2\check{\mathbf{G}}_0\mathbf{G}_0\right)\right]>c>0, \tag{53}$$

for all n, a condition that does not depend on the unknown form of $\boldsymbol{\Sigma}_0$. That is, the degree of heteroskedasticity does not affect consistency of the estimator.[12]

Third, if the errors are homoskedastic, it is immediate from (39) that $\boldsymbol{\psi}_0$ is identified if $n^{-1}\mathbf{Q}_0^{\prime}\mathbf{Q}_0$ is positive definite, and/or

$$n^{-1}Tr\left(\mathbf{G}_0^2+\mathbf{G}_0^{\prime}\mathbf{G}_0\right)-2n^{-2}\left[Tr(\mathbf{G}_0)\right]^2>c>0, \tag{54}$$

[11] Note that this moment condition is valid even if the pattern of heteroskedasticity is correlated with the spatial weights matrix.

[12] To see that (53) implies Assumption 5(b), let $\boldsymbol{A}_0=\left(a_{ij,0}\right)=\mathbf{G}_0^2+\mathbf{G}_0^{\prime}\mathbf{G}_0-2\check{\mathbf{G}}_0\mathbf{G}_0$ and $\mathbf{G}_0=\left(g_{ij,0}\right)$. It is easy to verify that $a_{ii,0}=\sum_{j=1,j\neq i}^{n}\left(g_{ij,0}g_{ji,0}+g_{ji,0}^2\right)$. But under Assumption 4 and $\rho_0\geq 0$, we have $g_{ij,0}\geq 0$, for all i and j, and then $Tr\left(n^{-1}\boldsymbol{A}_0\Sigma_0\right)=\sum_{i=1}^{n}\sigma_{i0}^2a_{ii,0}\geq\inf_i\left(\sigma_{i0}^2\right)Tr\left(n^{-1}\boldsymbol{A}_0\right)$. Therefore, $Tr\left(n^{-1}\boldsymbol{A}_0\right)>c>0$ implies $Tr\left(n^{-1}\boldsymbol{A}_0\Sigma_0\right)>c>0$.

for all n (including $n \to \infty$). Also notice that (54) is implied by (53) because

$$Tr\left(\check{\mathbf{G}}_0\mathbf{G}_0\right) = \sum_{i=1}^{n} g_{ii,0}^2 \geq \frac{1}{n}\left(\sum_{i=1}^{n} g_{ii,0}\right)^2 = \frac{1}{n}\left[Tr\left(\mathbf{G}_0\right)\right]^2.$$

Therefore, if (53) holds and $\rho_0 \geq 0$, identification is achieved irrespective of whether the errors are homoskedastic or not.

Interestingly, it turns out that the BMM estimator is related to the best GMM estimator under IID normal errors. The following proposition summarizes this relationship. Its proof is given in Section S1.2 of the Online Supplement.

Proposition 2. *Consider the SAR model given by (1), and assume that the errors are independently and normally distributed as $\varepsilon_i \sim IIDN\left(0, \sigma^2\right)$, for $i = 1, 2, \ldots, n$, and $0 < c < \sigma^2 < K$. Suppose that Assumptions 2–5 hold and the network centrality, δ, defined by (4), lies in the range $0 \leq \delta < 1/2$. Then the BMM estimator of $\boldsymbol{\psi}_0 = \left(\rho_0, \boldsymbol{\beta}_0'\right)'$, defined by (38), has the same asymptotic distribution as the best GMM estimator of $\boldsymbol{\psi}_0$, defined by (31) using $\mathbf{G}_0 - n^{-1}Tr\left(\mathbf{G}_0\right)\mathbf{I}_n$ in the quadratic moment condition, and $\left(\mathbf{G}_0\mathbf{X}\boldsymbol{\beta}_0, \mathbf{X}\right)$ in the linear moment conditions, where $\mathbf{G}_0$ is defined by (8).*

We complete the discussion of our theoretical results with the following two examples.

Example 1. Consider the following star network:

$$\mathbf{W} = \begin{pmatrix} 0 & w_{12} & 0 & 0 & \ldots & 0 & 0 \\ w_{21} & 0 & w_{23} & 0 & \ldots & 0 & 0 \\ w_{31} & 0 & 0 & w_{34} & \ldots & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ w_{n-1,1} & 0 & 0 & 0 & \ldots & 0 & w_{n-1,n} \\ w_{n1} & 0 & 0 & 0 & \ldots & 0 & 0 \end{pmatrix}, \tag{55}$$

where, without loss of generality, the first unit is the star (dominant unit). As established in Lemma S.11 of the Online Supplement, a necessary condition for (54) is given by $n^{-1}Tr\left(\mathbf{W}'\mathbf{W}\right) > c > 0$, for all n (including $n \to \infty$). Note that this condition is necessary for (54), irrespective of whether there exist dominant units (with $\delta > 0$) or not, and is also necessary for (53) under heteroskedasticity and $\rho_0 \geq 0$, as discussed above. For this example it is easily seen that

$$n^{-1}Tr\left(\mathbf{W}'\mathbf{W}\right) = n^{-1}\sum_{i=2}^{n} w_{i1}^2 + n^{-1}\sum_{i=1}^{n-1} w_{i,i+1}^2. \tag{56}$$

The first term refers to the strength of the dominant unit (the first unit), and the second term captures the strength of the connections of non-dominant units to the dominant unit. In the absence of regressors, identification of ρ_0 requires at least one of the two terms of (56) to tend to a non-zero value as $n \to \infty$. In the case where there is no dominant unit or the strength of the dominant unit is weak, namely $n^{-1}\sum_{i=2}^{n} w_{i1}^2 \to 0$, as $n \to \infty$, then identification requires $\lim_{n\to\infty} n^{-1}\sum_{i=1}^{n-1} w_{i,i+1}^2 > c > 0$. This condition is met if $\inf_i \left|w_{i,i+1}\right| > c$, namely if the dominant unit is impacted by almost all other units in the network. The $\mathbf{W}$ matrix in our Monte Carlo experiments follows a similar structure as (55).

Example 2. Although the current paper considers asymmetric $\mathbf{W}$ to allow for the presence of dominant units, the discussions on identification apply equally to symmetric $\mathbf{W}$. Let us first consider a simple example where there is only one social group in which everyone is connected to one another, and $w_{ii} = 0$. In this case, the matrix of connections is represented by $\mathbf{W} = (n-1)^{-1}\left(\mathbf{1}_n\mathbf{1}_n' - \mathbf{I}_n\right)$. It is easily verified that $n^{-1}Tr\left(\mathbf{W}^2\right) = 1/(n-1)$, which tends to zero as $n \to \infty$. Therefore, the necessary condition for identification, namely, $n^{-1}Tr\left(\mathbf{W}'\mathbf{W}\right) > c > 0$ for all n (including $n \to \infty$), is violated and the endogenous social effect is unidentifiable without exogenous regressors. Now suppose that there are R groups with n_r units in the rth group, for $r = 1, 2, \ldots, R$. Clearly, $\sum_{r=1}^{R} n_r = n$. The standard linear-in-means social interaction model assumes that individuals within a group have the same pairwise dependence, whereas individuals across different groups are not dependent. See Case (1991, 1992) for examples of empirical studies employing such a network structure. Then the matrix of group interactions, $\mathbf{W}$, can be represented by the following block diagonal matrix:

$$\mathbf{W} = Diag\left(\mathbf{W}_1, \mathbf{W}_2, \ldots, \mathbf{W}_R\right), \quad \mathbf{W}_r = \frac{1}{n_r - 1}\left(\mathbf{1}_{n_r}\mathbf{1}_{n_r}' - \mathbf{I}_{n_r}\right), \ r = 1, 2, \ldots, R.$$

Since we have shown that $Tr\left(\mathbf{W}_r^2\right) = n_r/(n_r - 1)$, it follows that

$$n^{-1}Tr\left(\mathbf{W}^2\right) = n^{-1}\sum_{r=1}^{R} Tr\left(\mathbf{W}_r^2\right) = \sum_{r=1}^{R}\left(\frac{1}{n_r - 1}\right)\pi_r,$$

where $\pi_r = n_r/n$ is the fraction of population in the rth group. Suppose that n_r rises with n such that $\pi_r \geq 0$, as $n \to \infty$. If R is fixed, then $\lim_{n\to\infty} n^{-1}Tr\left(\mathbf{W}^2\right) = 0$ and the group interaction effect is unidentified in the absence of exogenous explanatory variables.

6. Monte Carlo experiments

We now examine the small sample properties of the GMM and BMM estimators for SAR models with dominant units using Monte Carlo techniques. The Data Generating Process (DGP) is specified as follows:

$$y_i = \alpha + \rho y_i^* + \beta x_i + \varepsilon_i, \ \ i = 1, 2, \ldots, n, \tag{57}$$

where $y_i^* = \mathbf{w}_{i.,y}^{'}\mathbf{y}$, $\mathbf{y} = (y_1, y_2, \ldots, y_n)'$, and $\mathbf{w}_{i.,y}^{'}$ is the ith row of $\mathbf{W}_y$. The exogenous regressor, x_i, is also generated as a spatially correlated process:

$$x_i = \lambda x_i^* + v_i, \quad i = 1, 2, \ldots, n, \tag{58}$$

where $x_i^* = \mathbf{w}_{i.x}^{'}\mathbf{x}$, $\mathbf{x} = (x_1, x_2, \ldots, x_n)'$, and $\mathbf{w}_{i.,x}^{'}$ is the ith row of $\mathbf{W}_x$. Note that the spatial coefficients and weights matrices could be different for the $\mathbf{y}$ and $\mathbf{x}$ processes.

In matrix form, (57) can be rewritten as

$$\mathbf{y} = \mathbf{S}_y^{-1}(\rho)(\beta\mathbf{x} + \alpha\mathbf{1}_n) + \mathbf{u},$$

where $\mathbf{S}_y(\rho) = \mathbf{I}_n - \rho\mathbf{W}_y$, $\mathbf{u} = \mathbf{S}_y^{-1}(\rho)\boldsymbol{\varepsilon}$, $\boldsymbol{\varepsilon} = (\varepsilon_1, \varepsilon_2, \ldots, \varepsilon_n)'$, and $\mathbf{u} = (u_1, u_2, \ldots, u_n)'$. Similarly, (58) can be rewritten as $\mathbf{x} = \mathbf{S}_x^{-1}(\lambda)\boldsymbol{v}$, where $\mathbf{S}_x(\lambda) = \mathbf{I}_n - \lambda\mathbf{W}_x$ and $\boldsymbol{v} = (v_1, v_2, \ldots, v_n)'$. For the idiosyncratic errors, we consider both Gaussian and non-Gaussian processes:

- Gaussian errors: $\varepsilon_i \sim IIDN\left(0, \sigma_i^2\right)$ and $v_i \sim IIDN\left(0, \sigma_{v,i}^2\right)$.
- Non-Gaussian errors: $\varepsilon_i/\sigma_i \sim IID\left[\chi^2(2) - 2\right]/2$ and $v_i/\sigma_{v,i} \sim IID\left[\chi^2(2) - 2\right]/2$, where $\chi^2(2)$ denotes a chi-square random variable with two degrees of freedom.

The error variances are generated as $\sigma_i^2 = \sigma_\varepsilon^2\tau_{\varepsilon,i}^2$ and $\sigma_{v,i}^2 = \sigma_v^2\tau_{v,i}^2$. Then $Var(\boldsymbol{\varepsilon}) = \sigma_\varepsilon^2\boldsymbol{D}_\varepsilon$, where $\boldsymbol{D}_\varepsilon = Diag\left(\tau_{\varepsilon,1}^2, \tau_{\varepsilon,2}^2, \ldots, \tau_{\varepsilon,n}^2\right)$, and $Var(\boldsymbol{v}) = \sigma_v^2\boldsymbol{D}_v$, where $\boldsymbol{D}_v = Diag\left(\tau_{v,1}^2, \tau_{v,2}^2, \ldots, \tau_{v,n}^2\right)$. We consider both homoskedastic and heteroskedastic errors:

- Homoskedastic errors: $\tau_{\varepsilon,i}^2 = \tau_{v,i}^2 = 1$, for all i. Note that in this case $\sigma_i^2 = \sigma_\varepsilon^2$ and $\sigma_{v,i}^2 = \sigma_v^2$, for all i, and $\boldsymbol{D}_\varepsilon = \boldsymbol{D}_v = \mathbf{I}_n$.
- Heteroskedastic errors: $\tau_{\varepsilon,i}^2 \sim IIDU(0.5, 1.5)$ and $\tau_{v,i}^2 \sim IIDU(0.5, 1.5)$, where $U(a, b)$ denotes the uniform distribution on the interval (a, b).[13]

When $\beta = 0$, the average fit of the SAR model is given by

$$R_0^2 = 1 - \frac{Tr\left[Var(\boldsymbol{\varepsilon})\right]}{Tr\left[Var(\mathbf{y})\right]} = 1 - \frac{Tr\left(\boldsymbol{D}_\varepsilon\right)}{Tr\left[\mathbf{S}_y^{-1}(\rho)\boldsymbol{D}_\varepsilon\mathbf{S}_y^{\prime-1}(\rho)\right]}, \tag{59}$$

which is determined by the choice of ρ and $\mathbf{W}$. To control the average fit of the SAR model when $\beta \neq 0$, we note that

$$R_\beta^2 = 1 - \frac{Tr\left[Var(\boldsymbol{\varepsilon})\right]}{Tr\left[Var(\mathbf{y})\right]}, \tag{60}$$

where

$$Tr\left[Var(\mathbf{y})\right] = \beta^2\sigma_v^2 Tr\left[\mathbf{S}_y^{-1}(\rho)\mathbf{S}_x^{-1}(\lambda)\boldsymbol{D}_v\mathbf{S}_x^{\prime-1}(\lambda)\mathbf{S}_y^{\prime-1}(\rho)\right] + \sigma_\varepsilon^2 Tr\left[\mathbf{S}_y^{-1}(\rho)\boldsymbol{D}_\varepsilon\mathbf{S}_y^{\prime-1}(\rho)\right].$$

It is also easily seen that

$$R_\beta^2 - R_0^2 = \frac{a_n s^2\left(1 - R_0^2\right)}{1 + a_n s^2} \geq 0,$$

where

$$a_n = \frac{Tr\left[\mathbf{S}_y^{-1}(\rho)\mathbf{S}_x^{-1}(\lambda)\boldsymbol{D}_v\mathbf{S}_x^{\prime-1}(\lambda)\mathbf{S}_y^{\prime-1}(\rho)\right]}{Tr\left[\mathbf{S}_y^{-1}(\rho)\boldsymbol{D}_\varepsilon\mathbf{S}_y^{\prime-1}(\rho)\right]} > 0, \ s^2 = \frac{\beta^2\sigma_v^2}{\sigma_\varepsilon^2} \geq 0, \tag{61}$$

and note that s^2 is the signal-to-noise ratio. Since $a_n s^2 \geq 0$, we have $R_\beta^2 \geq R_0^2$, with equality holding if and only if $\beta = 0$. Therefore, given the values of $\mathbf{W}_y$ and ρ we can only control the value of $R_\beta^2 - R_0^2$. Since we are interested in the effects of changes in ρ and δ on the property of GMM and BMM estimators, without loss of generality we set the other parameter

[13] We also carried out Monte Carlo experiments allowing heteroskedasticity of the error variances, $\tau_{\varepsilon,i}^2$ and $\tau_{v,i}^2$, to be correlated with the spatial weights, w_{ij}. Both GMM and BMM estimators are found to work well, and seem to be robust to such a correlation. The results of these experiments are available upon request.

Table 1a
Bias and RMSE of the GMM and BMM estimators of ρ for the experiments with homoskedastic non-Gaussian errors.

	GMM[b]								BMM							
	Bias(×100)				RMSE(×100)				Bias(×100)				RMSE(×100)			
δ	n															
	100	300	500	1000	100	300	500	1000	100	300	500	1000	100	300	500	1000
	$\rho_0 = 0.2$															
0.00	−6.09	−1.87	−1.18	−0.59	19.86	9.83	7.55	5.28	−8.18	−2.39	−1.47	−0.73	20.28	9.90	7.59	5.30
0.25	−6.03	−1.85	−1.17	−0.59	19.83	9.82	7.55	5.28	−8.10	−2.37	−1.46	−0.73	20.24	9.88	7.59	5.30
0.50	−6.21	−1.89	−1.21	−0.60	19.84	9.94	7.58	5.29	−8.30	−2.42	−1.49	−0.74	20.26	9.99	7.62	5.31
0.75	−5.99	−1.84	−1.25	−0.60	20.96	10.35	7.92	5.49	−8.70	−2.52	−1.60	−0.76	21.18	10.38	7.94	5.50
0.95	−4.80	−1.60	−1.17	−0.59	27.26	13.04	9.70	6.68	−10.82	−3.13	−1.97	−0.96	25.51	12.62	9.52	6.58
1.00	−4.57	−1.37	−1.01	−0.55	31.80	15.63	11.78	8.07	−11.95	−3.50	−2.19	−1.10	28.55	14.55	11.28	7.84
	$\rho_0 = 0.5$															
0.00	−5.28	−1.62	−1.01	−0.50	15.88	7.40	5.61	3.90	−7.36	−2.11	−1.27	−0.63	16.39	7.49	5.66	3.92
0.25	−5.27	−1.61	−1.01	−0.51	15.89	7.39	5.61	3.90	−7.32	−2.10	−1.27	−0.63	16.38	7.48	5.65	3.92
0.50	−5.43	−1.65	−1.03	−0.52	15.96	7.48	5.63	3.91	−7.49	−2.14	−1.29	−0.64	16.43	7.57	5.68	3.94
0.75	−5.33	−1.70	−1.16	−0.56	17.34	7.99	6.06	4.14	−8.14	−2.34	−1.48	−0.71	17.60	8.07	6.10	4.16
0.95	−4.06	−1.55	−1.18	−0.59	24.92	11.14	8.14	5.39	−11.57	−3.45	−2.13	−1.03	22.97	10.65	7.88	5.32
1.00	−4.10	−0.81	−0.75	−0.46	29.49	14.60	10.70	7.07	−13.67	−4.10	−2.53	−1.25	26.63	12.80	9.80	6.73
	$\rho_0 = 0.75$															
0.00	−3.80	−1.17	−0.72	−0.36	10.68	4.61	3.41	2.35	−5.57	−1.55	−0.92	−0.45	11.18	4.69	3.46	2.37
0.25	−3.81	−1.17	−0.72	−0.36	10.71	4.60	3.41	2.35	−5.56	−1.55	−0.92	−0.46	11.19	4.69	3.46	2.37
0.50	−3.87	−1.19	−0.74	−0.37	10.83	4.64	3.42	2.37	−5.65	−1.57	−0.93	−0.46	11.25	4.72	3.47	2.39
0.75	−3.91	−1.30	−0.88	−0.42	12.49	5.15	3.84	2.57	−6.46	−1.82	−1.14	−0.54	12.59	5.24	3.88	2.59
0.95	−4.04	−0.94	−0.87	−0.46	22.04	9.04	6.15	3.80	−11.54	−3.38	−2.04	−0.97	19.40	7.96	5.65	3.67
1.00	−6.09	−0.10	0.22	0.13	27.25	14.06	10.46	6.97	−15.44	−4.97	−3.05	−1.51	24.28	10.80	8.10	5.49

Notes: The data generating process (DGP) is given by (57) and (58) with homoskedastic non-Gaussian errors. $\mathbf{W}_x = \mathbf{W}_y = \mathbf{W}$. The first unit is δ-dominant, and the rest of the units are non-dominant. The number of replications is 2000. The BMM estimator is computed by (38).
[b] The GMM estimator refers to the best GMM and is computed in two steps: In the first step, we obtain preliminary GMM estimates, $\tilde{\boldsymbol{\psi}} = \left(\tilde{\rho}, \tilde{\alpha}, \tilde{\beta}\right)'$, following (23) with equal weights, where $\mathbf{Z} = \left(\mathbf{1}_n, \mathbf{x}, \mathbf{W}\mathbf{x}, \mathbf{W}^2\mathbf{x}\right)$, $\mathbf{B}_1 = \mathbf{W}$, $\mathbf{B}_2 = \mathbf{W}^2 - n^{-1}Tr\left(\mathbf{W}^2\right)\mathbf{I}_n$. In the second step, we use $\left(\tilde{\mathbf{G}}\mathbf{x}\tilde{\alpha}, \tilde{\mathbf{G}}\mathbf{x}\tilde{\beta}, \mathbf{1}_n, \mathbf{x}\right)$ and $\tilde{\mathbf{G}} - n^{-1}Tr\left(\tilde{\mathbf{G}}\right)\mathbf{I}_n$, where $\tilde{\mathbf{G}} = \mathbf{W}\left(\mathbf{I}_n - \tilde{\rho}\mathbf{W}\right)^{-1}$, in the linear and quadratic moments, respectively, and compute the optimal GMM estimates by (31).

values to $\sigma^2_{\varepsilon,0} = 1$, $\lambda_0 = 0.75$, $\alpha_0 = 1$ and $\beta_0 = 1$. The value of σ_v^2 is chosen to ensure that $R^2_\beta = R^2_0 + 0.1$. This is achieved by setting σ_v^2 such that

$$\frac{\beta^2\sigma_v^2}{\sigma_\varepsilon^2} \times \frac{Tr\left[\mathbf{S}_y^{-1}(\rho)\,\mathbf{S}_x^{-1}(\lambda)\,\boldsymbol{D}_v\mathbf{S}_x'^{-1}(\lambda)\,\mathbf{S}_y'^{-1}(\rho)\right]}{Tr\left[\mathbf{S}_y^{-1}(\rho)\,\boldsymbol{D}_\varepsilon\mathbf{S}_y'^{-1}(\rho)\right]} = \frac{0.1}{0.9 - R_0^2},$$

or equivalently,

$$\sigma_v^2 = \left(\frac{0.1}{0.9 - R_0^2}\right)\frac{\sigma_\varepsilon^2}{\beta^2 a_n}, \tag{62}$$

where a_n is defined by (61). The value of ρ is now chosen so that $R_0^2 < 0.9$.

Turning to the specifications of the spatial weights matrices, we consider the case where $\mathbf{W}_x = \mathbf{W}_y = \mathbf{W}$ in the main text and report the results for the choices with $\mathbf{W}_x \neq \mathbf{W}_y$ in the Online Supplement. The spatial weights matrix $\mathbf{W}$,

$$\mathbf{W} = (w_{ij})_{n\times n} = \begin{pmatrix} 0 & \mathbf{w}_{12}' \\ \mathbf{w}_{21} & \mathbf{W}_{22} \end{pmatrix},$$

is generated as follows: We assume, without loss of generality, that the first unit of the network is δ-dominant and the rest are non-dominant. Specifically, the first $\lfloor n^\delta \rfloor$ elements of the $(n-1)\times 1$ column vector $\mathbf{w}_{21}$ are drawn from $IIDU(0, 1)$ and the rest are set to zero, where $\lfloor . \rfloor$ is the integer part operator. In this way, the sum of the first column of $\mathbf{W}$ expands with n at the rate of δ, i.e., $\sum_{i=1}^n w_{i1} = \ominus\left(n^\delta\right)$. The first 8 elements of the $1\times(n-1)$ row vector $\mathbf{w}_{12}'$ are set to one and the remaining elements to zero. $\mathbf{W}_{22}$ is a standard $(n-1)\times(n-1)$ spatial matrix with 8 connections (4-ahead-and-4-behind with equal weights), namely, $w_{i,j} = 0.125$ for $j = i-4, \ldots, i-1, i+1, \ldots, i+4$, and $w_{i,j} = 0$ otherwise. By construction, $\mathbf{W}_{22}$ is uniformly bounded in both row and column norms, namely, $\|\mathbf{W}_{22}\|_1 = O(1)$ and $\|\mathbf{W}_{22}\|_\infty = O(1)$. Finally, $\mathbf{W}$ is standardized so that each row sums to one.[14]

[14] Row standardization is unnecessary and is imposed for convenience. Recall that our theory only requires the row sums of $\mathbf{W}$ to be uniformly bounded.

Table 1b
Size and Power of the GMM and BMM estimators of ρ for the experiments with homoskedastic non-Gaussian errors.

	GMM[b]								BMM							
	Size($\times$100)				Power($\times$100)				Size($\times$100)				Power($\times$100)			
δ	n															
	100	300	500	1000	100	300	500	1000	100	300	500	1000	100	300	500	1000
	$\rho_0 = 0.2$															
0.00	7.45	4.30	5.30	5.20	9.85	16.30	24.25	43.15	6.60	3.95	5.20	5.20	7.80	14.85	22.70	42.30
0.25	7.45	4.30	5.35	5.25	9.65	16.15	24.20	43.20	6.90	4.00	5.25	5.25	7.65	14.80	22.75	42.35
0.50	7.40	4.45	5.20	5.15	9.85	16.60	24.50	42.90	6.95	4.35	5.25	5.10	7.45	15.05	22.95	42.30
0.75	9.05	5.45	6.15	6.65	11.70	18.45	24.95	42.75	8.15	5.20	6.15	6.40	8.35	16.35	23.60	41.95
0.95	18.70	12.55	12.00	10.00	21.75	23.25	26.85	41.60	13.30	9.85	10.35	9.45	12.95	18.65	23.75	40.00
1.00	24.70	17.55	17.45	16.05	25.45	26.20	29.80	41.35	16.85	13.75	15.35	13.80	16.45	21.25	25.85	38.45
	$\rho_0 = 0.5$															
0.00	7.70	4.40	5.25	5.65	14.00	25.80	39.40	68.05	6.60	4.35	4.90	5.50	8.95	23.80	37.50	67.10
0.25	7.95	4.50	5.25	5.55	14.00	25.95	39.25	68.15	6.85	4.35	4.90	5.50	9.05	23.65	37.50	67.25
0.50	8.00	4.75	5.20	5.80	14.40	25.95	38.95	67.90	6.75	4.45	5.30	5.50	9.15	24.20	37.15	66.85
0.75	11.00	5.75	6.95	6.40	16.95	26.65	38.20	65.85	7.95	5.60	6.45	6.10	9.85	23.75	36.25	64.10
0.95	25.80	16.45	14.50	12.15	28.65	32.15	37.75	57.60	14.55	11.85	12.25	10.45	14.45	23.95	32.65	54.85
1.00	32.90	25.30	23.55	22.15	33.05	34.80	39.80	55.95	20.35	18.45	19.60	19.60	19.05	27.10	33.85	51.80
	$\rho_0 = 0.75$															
0.00	9.55	4.75	4.95	5.75	25.80	56.40	77.25	96.55	6.65	4.30	4.95	5.85	18.40	52.65	75.45	96.20
0.25	9.85	4.75	4.85	5.50	25.85	56.50	77.45	96.45	6.60	4.35	4.75	5.55	18.65	52.65	75.55	96.10
0.50	10.35	4.95	5.25	5.50	26.05	56.50	77.30	96.60	6.95	4.65	4.80	5.65	18.85	53.00	75.60	96.35
0.75	14.85	7.00	7.70	6.80	29.30	54.80	73.50	94.45	8.55	6.40	7.15	6.75	18.55	49.45	70.80	94.25
0.95	40.75	26.00	23.05	17.20	42.00	51.15	62.70	86.15	17.75	14.90	15.50	13.25	19.15	39.20	56.10	84.30
1.00	52.40	43.00	42.95	38.70	45.85	49.25	59.60	75.85	27.40	30.75	31.40	32.10	23.35	37.00	49.45	72.10

Notes: The power is calculated at $\rho = \rho_0 - 0.1$, where ρ_0 denotes the true value. See also the notes to Table 1a.

We consider a number of different values of δ and ρ_0: $\delta = 0, 0.25, 0.50, 0.75, 0.95, 1$, and $\rho_0 = 0.2, 0.5, 0.75$;[15] and experiment with four sample sizes: $n = 100, 300, 500$, and 1000. We include $\delta = 1$ in our experiments in order to see if the GMM and BMM estimators break down when $\delta = 1$, as predicted by the theory, and to examine how the two estimators perform as δ approaches unity. The number of replications is set to 2,000, per experiment. We report results for both GMM and BMM estimates. The BMM estimator assuming homoskedastic errors is computed by (38), and by (48) under heteroskedastic errors. The GMM estimator reported here is the best GMM estimator.[16] Specifically, the GMM estimator assuming homoskedastic errors is computed in two steps: In the first step, the GMM estimates are computed with equal weights using $\mathbf{B}_1 = \mathbf{W}$, $\mathbf{B}_2 = \mathbf{W}^2 - n^{-1}Tr\left(\mathbf{W}^2\right)\mathbf{I}_n$, and $\mathbf{Z} = \left(\mathbf{1}_n, \mathbf{x}, \mathbf{Wx}, \mathbf{W}^2\mathbf{x}\right)$. In the second step, we re-estimate with the optimal GMM weights using the best IV matrix, $\left(\tilde{\mathbf{G}}\mathbf{x}\tilde{\alpha}, \tilde{\mathbf{G}}\mathbf{x}\tilde{\boldsymbol{\beta}}, \mathbf{1}_n, \mathbf{x}\right)$, and $\tilde{\mathbf{G}} - n^{-1}Tr\left(\tilde{\mathbf{G}}\right)\mathbf{I}_n$ in the quadratic moment, where $\tilde{\mathbf{G}} = \mathbf{G}(\tilde{\rho})$, and $\left(\tilde{\rho}, \tilde{\alpha}, \tilde{\beta}\right)'$ denote the first-step GMM estimates. Similarly, the GMM estimator assuming heteroskedastic errors is obtained in two steps, but it uses different matrices in the quadratic moments, namely, $\mathbf{B}_1 = \mathbf{W}$ and $\mathbf{B}_2 = \mathbf{W}^2 - Diag\,(\mathbf{W})$ in the first step, and $\tilde{\mathbf{G}} - Diag\left(\tilde{\mathbf{G}}\right)$ in the second step.

Tables 1a, 1b, 2a and 2b summarize the results for homoskedastic non-Gaussian errors, and Tables 3a, 3b, 4a and 4b report the results for heteroskedastic non-Gaussian errors.[17] For each experiment, we report bias, root mean square error (RMSE), size, and power of both estimators for ρ and β. The results for the intercept term are not reported to save space. In addition, Figs. 1a, 1b, 2a and 2b plot the empirical power functions for ρ and β in the case of $\rho_0 = 0.5$ and $\beta_0 = 1$ for $\delta = 0, 0.25, 0.75, 0.95$, and $n = 100$ and 300, when the errors are homoskedastic non-Gaussian.[18]

Let us begin by examining the bias and RMSE results. We first observe from Tables 1a and 2a that both GMM and BMM estimators display declining bias and RMSE as the sample size increases. On the whole, the bias and RMSE are very small even when $n = 100$, irrespective of the magnitude of the spatial autoregressive parameter, ρ. This result is in line with our theoretical finding that both estimators are consistent if $\delta < 1$. However, as the value of δ approaches one, we see a substantial increase in RMSE for both estimators. The two estimators perform similarly in terms of RMSE when $n > 300$, although the BMM estimator of ρ has smaller RMSE than the GMM estimator when $n = 100$ and δ is close to unity, despite being more biased. The performance of the two estimators are quite close when we consider β, giving very similar RMSEs for all sample sizes under consideration. Finally, the bias and RMSE of both methods are robust to

[15] The values of R_0^2 for different ρ and δ are reported in Eq. (59) in the Online Supplement. Note that $R_0^2 < 0.9$ holds when $\rho \leq 0.75$. When $\beta \neq 0$, we set $R_\beta^2 = R_0^2 + 0.1$. We have also examined the estimation of SAR models without exogenous regressors ($\beta = 0$), with the results given in the Online Supplement.

[16] We also consider GMM estimators using other IV and $\mathbf{B}$ matrices. The results are presented in Section 2.1 of the Online Supplement.

[17] Monte Carlo results for the case of Gaussian errors are provided in the Online Supplement to save space.

[18] Power function plots when errors are heteroskedastic, and for other values of ρ_0, δ and n are presented in the Online Supplement.

Table 2a
Bias and RMSE of the GMM and BMM estimators of β for the experiments with homoskedastic non-Gaussian errors.

	GMM[b]								BMM							
	Bias($\times 100$)				RMSE($\times 100$)				Bias($\times 100$)				RMSE($\times 100$)			
δ	n															
	100	300	500	1000	100	300	500	1000	100	300	500	1000	100	300	500	1000
	$\rho_0 = 0.2$															
0.00	1.30	0.43	0.74	0.24	39.71	21.06	15.91	11.32	4.68	1.43	1.30	0.53	40.54	21.26	16.04	11.36
0.25	1.27	0.40	0.73	0.24	39.67	21.05	15.89	11.31	4.63	1.40	1.29	0.53	40.51	21.25	16.02	11.35
0.50	1.29	0.33	0.71	0.22	39.77	21.11	15.96	11.33	4.67	1.36	1.27	0.50	40.60	21.32	16.08	11.38
0.75	0.80	0.37	0.74	0.29	41.20	21.77	16.50	11.79	4.41	1.52	1.35	0.60	41.90	21.95	16.62	11.82
0.95	0.45	−0.01	0.65	0.16	47.26	26.83	20.70	16.07	4.11	1.42	1.45	0.59	47.68	26.97	20.83	16.09
1.00	0.16	−0.08	0.68	0.08	51.74	31.10	25.58	21.66	3.52	1.16	1.42	0.44	52.29	31.33	25.75	21.70
	$\rho_0 = 0.5$															
0.00	1.83	0.65	0.95	0.33	47.05	24.73	18.63	13.23	6.26	1.91	1.66	0.69	48.26	25.02	18.81	13.30
0.25	1.81	0.62	0.94	0.33	47.00	24.72	18.61	13.22	6.22	1.89	1.65	0.69	48.20	25.00	18.79	13.28
0.50	1.84	0.54	0.92	0.30	47.25	24.92	18.77	13.33	6.29	1.85	1.63	0.66	48.48	25.24	18.97	13.39
0.75	1.38	0.66	1.07	0.45	51.53	27.83	21.36	15.75	6.23	2.16	1.87	0.85	52.61	28.17	21.57	15.80
0.95	1.55	0.43	1.56	0.55	71.50	48.66	40.72	35.96	6.55	2.52	2.70	1.19	72.66	49.25	41.18	36.07
1.00	1.48	0.42	2.02	0.48	85.79	64.95	59.45	56.96	5.97	2.34	3.15	1.03	87.77	65.91	60.20	57.22
	$\rho_0 = 0.75$															
0.00	2.33	0.84	1.09	0.40	49.36	25.64	19.22	13.65	7.89	2.43	1.99	0.85	50.77	25.98	19.45	13.72
0.25	2.33	0.83	1.09	0.40	49.30	25.62	19.20	13.63	7.87	2.41	1.98	0.85	50.70	25.95	19.42	13.70
0.50	2.31	0.73	1.07	0.38	49.61	26.00	19.50	13.87	7.99	2.38	1.97	0.82	51.06	26.37	19.74	13.94
0.75	2.19	0.96	1.38	0.62	58.15	32.64	25.54	19.54	8.35	2.91	2.45	1.15	59.66	33.12	25.85	19.61
0.95	3.44	0.91	2.63	0.97	96.55	73.12	63.65	58.72	9.66	4.05	4.26	1.97	99.21	74.27	64.60	58.90
1.00	3.63	0.78	3.13	0.65	120.03	101.13	95.82	95.44	8.90	3.87	5.28	1.87	124.04	103.80	98.10	96.30

Notes: The true parameter value is $\beta_0 = 1$. See also the notes to Table 1a.

Table 2b
Size and power of the GMM and BMM estimators of β for the experiments with homoskedastic non-Gaussian errors.

	GMM[b]								BMM							
	Size($\times 100$)				Power($\times 100$)				Size($\times 100$)				Power($\times 100$)			
δ	n															
	100	300	500	1000	100	300	500	1000	100	300	500	1000	100	300	500	1000
	$\rho_0 = 0.2$															
0.00	6.70	5.05	5.60	5.95	10.45	17.05	25.15	44.40	6.85	5.45	5.85	6.15	11.90	18.85	26.65	45.60
0.25	6.70	5.00	5.50	5.90	10.30	16.95	25.15	44.35	6.90	5.35	5.70	6.15	11.80	18.60	26.85	45.60
0.50	6.45	5.55	5.45	5.65	9.65	16.85	26.15	44.25	6.60	5.75	5.60	5.75	11.20	18.35	27.15	45.05
0.75	6.80	5.00	5.05	6.00	10.15	16.00	24.40	41.75	6.80	5.20	5.20	6.00	11.60	17.55	25.80	42.60
0.95	6.70	6.05	5.95	5.70	9.30	14.00	18.10	26.55	6.65	5.90	5.90	5.85	10.50	15.15	18.75	27.35
1.00	6.20	5.95	5.30	5.30	8.35	10.50	13.00	16.55	6.35	6.00	5.50	5.60	9.35	11.55	13.95	17.05
	$\rho_0 = 0.5$															
0.00	6.75	5.15	5.65	5.80	9.55	14.25	20.15	34.70	6.80	5.35	5.85	6.05	11.00	15.25	21.25	35.75
0.25	6.75	5.20	5.60	5.85	9.55	14.15	20.35	34.75	6.80	5.30	5.75	5.95	11.00	15.15	21.30	35.50
0.50	6.60	5.35	5.35	5.75	8.95	13.85	19.55	34.15	6.60	5.65	5.65	5.75	10.55	14.85	20.70	35.30
0.75	7.00	5.05	5.45	6.05	8.65	12.65	17.20	26.60	7.05	5.25	5.55	6.10	10.00	13.45	18.45	27.60
0.95	6.70	6.00	5.85	5.70	7.80	8.25	9.10	9.25	7.05	5.85	5.95	5.90	8.85	9.20	9.55	9.45
1.00	6.05	5.80	5.35	5.75	6.90	6.30	6.20	6.75	6.35	6.15	5.60	5.90	7.90	6.95	6.60	7.00
	$\rho_0 = 0.75$															
0.00	7.05	5.00	5.40	5.70	9.45	13.95	19.30	32.85	7.35	5.40	5.65	5.95	10.80	15.25	20.55	33.95
0.25	7.00	5.05	5.45	5.70	9.60	14.05	19.20	33.10	7.40	5.45	5.60	5.75	11.00	15.35	20.65	34.15
0.50	7.45	5.30	5.30	5.70	9.25	13.00	19.00	31.60	7.35	5.60	5.70	5.80	10.40	14.65	20.05	32.70
0.75	7.40	5.25	5.45	6.25	8.95	10.75	14.00	19.60	7.35	5.75	5.65	6.25	10.10	11.95	14.75	20.40
0.95	6.90	5.85	5.75	6.10	6.90	6.85	6.95	7.50	7.60	5.95	6.15	6.10	8.60	7.65	7.40	7.60
1.00	6.10	5.55	5.20	5.60	6.75	5.70	5.35	5.95	6.95	6.40	5.60	6.10	7.65	6.15	6.15	6.20

Notes: The true parameter value is $\beta_0 = 1$ and power is calculated at 0.8. See also the notes to Table 1a.

non-Gaussian errors.[19] Both estimators exhibit similar bias and RMSE properties under heteroskedasticity, as shown in Tables 3a and 4a.

[19] This can be seen by comparing the results in Tables 1a and 2a with the ones reported under homoskedastic Gaussian errors provided in Tables S.7 to S.10 of the Online Supplement.

Table 3a
Bias and RMSE of the GMM and BMM estimators of ρ for the experiments with heteroskedastic non-Gaussian errors.

	GMM[b]								BMM							
	Bias($\times 100$)				RMSE($\times 100$)				Bias($\times 100$)				RMSE($\times 100$)			
δ	n															
	100	300	500	1000	100	300	500	1000	100	300	500	1000	100	300	500	1000
	$\rho_0 = 0.2$															
0.00	−6.10	−1.88	−1.22	−0.62	19.92	9.89	7.64	5.30	−8.06	−2.33	−1.45	−0.73	20.19	9.87	7.63	5.30
0.25	−6.05	−1.86	−1.21	−0.62	19.87	9.87	7.64	5.30	−7.98	−2.31	−1.43	−0.73	20.13	9.85	7.62	5.29
0.50	−6.22	−1.91	−1.24	−0.63	19.88	9.97	7.66	5.30	−8.16	−2.35	−1.46	−0.73	20.15	9.95	7.65	5.30
0.75	−6.24	−1.90	−1.31	−0.63	21.20	10.41	7.97	5.49	−8.62	−2.46	−1.56	−0.75	21.21	10.38	7.94	5.48
0.95	−5.97	−1.93	−1.34	−0.65	27.05	12.96	9.58	6.50	−10.54	−3.04	−1.86	−0.91	25.45	12.58	9.38	6.42
1.00	−6.64	−2.06	−1.42	−0.75	31.29	15.49	11.65	7.91	−11.81	−3.39	−2.10	−1.07	28.83	14.56	11.20	7.69
	$\rho_0 = 0.5$															
0.00	−5.32	−1.61	−1.02	−0.52	15.92	7.43	5.68	3.92	−7.28	−2.06	−1.26	−0.63	16.30	7.44	5.68	3.92
0.25	−5.30	−1.61	−1.02	−0.52	15.91	7.42	5.68	3.92	−7.24	−2.05	−1.25	−0.63	16.26	7.43	5.68	3.92
0.50	−5.40	−1.64	−1.04	−0.53	15.95	7.47	5.70	3.93	−7.35	−2.08	−1.27	−0.64	16.28	7.49	5.70	3.93
0.75	−5.52	−1.69	−1.15	−0.55	17.32	7.95	6.03	4.10	−7.97	−2.25	−1.41	−0.68	17.42	7.97	6.02	4.10
0.95	−5.44	−1.85	−1.24	−0.61	24.11	10.83	7.77	5.15	−10.90	−3.18	−1.90	−0.92	22.45	10.35	7.53	5.06
1.00	−6.41	−1.87	−1.30	−0.71	28.99	14.31	10.62	7.05	−13.09	−3.90	−2.38	−1.19	26.50	12.73	9.68	6.59
	$\rho_0 = 0.75$															
0.00	−3.86	−1.17	−0.73	−0.37	10.73	4.61	3.46	2.37	−5.51	−1.52	−0.91	−0.46	11.08	4.64	3.48	2.38
0.25	−3.85	−1.17	−0.73	−0.37	10.75	4.61	3.46	2.37	−5.50	−1.52	−0.91	−0.46	11.06	4.64	3.47	2.38
0.50	−3.88	−1.18	−0.74	−0.38	10.79	4.61	3.47	2.38	−5.53	−1.52	−0.92	−0.46	11.06	4.65	3.48	2.38
0.75	−3.95	−1.27	−0.85	−0.40	12.30	5.02	3.75	2.51	−6.22	−1.72	−1.06	−0.50	12.21	5.06	3.76	2.51
0.95	−5.14	−1.37	−0.87	−0.46	21.12	8.29	5.68	3.55	−10.57	−2.99	−1.73	−0.82	18.53	7.50	5.21	3.40
1.00	−8.48	−2.15	−1.20	−0.59	28.88	14.05	10.29	6.89	−14.50	−4.66	−2.84	−1.44	23.89	10.77	8.03	5.40

Notes: The data generating process (DGP) is given by (57) and (58) with heteroskedastic non-Gaussian errors. $\mathbf{W}_x = \mathbf{W}_y = \mathbf{W}$. The first unit is δ-dominant, and the rest of the units are non-dominant. The number of replications is 2000. The BMM estimator is computed by (48). [b] The GMM estimator refers to the best GMM and is computed in two steps: In the first step, we obtain preliminary GMM estimates, $\tilde{\boldsymbol{\psi}} = \left(\tilde{\rho}, \tilde{\alpha}, \tilde{\beta}\right)'$, following (23) with equal weights, where $\mathbf{Z} = \left(\mathbf{1}_n, \mathbf{x}, \mathbf{W}\mathbf{x}, \mathbf{W}^2\mathbf{x}\right)$, $\mathbf{B}_1 = \mathbf{W}$, $\mathbf{B}_2 = \mathbf{W}^2 - Diag\left(\mathbf{W}^2\right)$. In the second step, we use $\left(\tilde{\mathbf{G}}\mathbf{x}\tilde{\alpha}, \tilde{\mathbf{G}}\mathbf{x}\tilde{\beta}, \mathbf{1}_n, \mathbf{x}\right)$ and $\tilde{\mathbf{G}} - Diag\left(\tilde{\mathbf{G}}\right)$, where $\tilde{\mathbf{G}} = \mathbf{W}\left(\mathbf{I}_n - \tilde{\rho}\mathbf{W}\right)^{-1}$, in the linear and quadratic moments, respectively, and compute the optimal GMM estimates by (31).

Table 3b
Size and Power of the GMM and BMM estimators of ρ for the experiments with heteroskedastic non-Gaussian errors.

	GMM[b]								BMM							
	Size($\times 100$)				Power($\times 100$)				Size($\times 100$)				Power($\times 100$)			
δ	n															
	100	300	500	1000	100	300	500	1000	100	300	500	1000	100	300	500	1000
	$\rho_0 = 0.2$															
0.00	10.75	5.90	6.20	5.35	11.45	15.90	24.20	43.35	9.55	5.55	5.75	5.10	8.35	13.95	22.95	42.30
0.25	10.50	6.10	6.20	5.40	11.30	15.75	24.10	43.50	9.50	5.65	5.70	5.10	8.25	14.15	22.90	42.40
0.50	9.95	6.20	5.90	6.00	11.15	16.10	24.20	42.85	9.30	5.95	5.70	5.65	9.00	14.35	23.10	42.25
0.75	11.05	6.85	6.80	5.95	12.20	17.15	24.20	43.00	9.90	6.20	6.20	5.75	9.10	14.60	22.45	41.65
0.95	17.00	11.15	10.05	8.25	17.80	19.40	22.90	37.95	12.80	8.85	8.60	7.60	11.60	15.65	20.80	36.85
1.00	20.55	14.40	14.15	11.85	21.60	19.80	23.20	33.95	15.45	12.05	12.25	10.65	14.55	16.35	20.90	32.35
	$\rho_0 = 0.5$															
0.00	10.40	6.10	6.15	5.95	15.10	25.80	40.80	68.65	9.30	5.70	5.60	5.55	10.40	23.40	39.30	68.15
0.25	10.50	6.10	6.20	5.80	15.30	25.50	41.00	68.55	9.10	5.60	5.55	5.55	10.50	23.20	39.30	68.15
0.50	11.05	6.05	6.05	5.75	14.95	26.65	40.20	68.80	9.50	6.00	6.00	5.75	10.60	23.75	38.85	67.60
0.75	12.30	6.90	7.30	6.15	16.55	26.10	39.05	66.15	10.10	6.80	6.85	6.35	10.45	23.50	37.10	65.30
0.95	21.85	13.85	11.90	9.30	24.45	26.75	35.45	56.15	14.35	10.10	9.30	8.20	13.50	21.30	31.45	53.85
1.00	27.85	20.00	18.95	15.85	27.40	27.45	31.50	45.45	18.75	15.75	15.75	13.50	17.30	21.50	27.65	44.35
	$\rho_0 = 0.75$															
0.00	13.35	5.90	6.20	5.60	27.20	58.30	79.25	97.05	10.15	5.45	5.75	5.40	20.35	54.55	77.80	96.65
0.25	13.40	6.10	6.25	5.55	27.40	58.35	79.15	97.15	10.40	5.50	5.70	5.50	20.90	54.75	77.85	96.80
0.50	13.30	5.85	5.95	5.60	27.60	57.90	78.95	97.05	10.15	5.60	5.35	5.60	20.40	54.70	77.55	96.65
0.75	15.35	6.60	7.50	6.30	29.50	55.75	75.05	96.15	10.75	6.15	6.80	6.25	19.90	51.85	73.65	95.75
0.95	33.75	20.00	16.10	12.80	36.55	45.80	60.10	85.85	17.05	12.85	12.35	10.45	18.25	37.40	55.85	84.55
1.00	44.05	33.95	32.55	31.65	39.75	40.10	46.40	63.45	25.45	22.90	24.75	24.35	22.50	32.10	44.20	63.65

Notes: The power is calculated at $\rho = \rho_0 - 0.1$, where ρ_0 denotes the true value. See also the notes to Table 3a.

Table 4a
Bias and RMSE of the GMM and BMM estimators of β for the experiments with heteroskedastic non-Gaussian errors.

	GMM[b]								BMM							
	Bias($\times 100$)				RMSE($\times 100$)				Bias($\times 100$)				RMSE($\times 100$)			
δ	n															
	100	300	500	1000	100	300	500	1000	100	300	500	1000	100	300	500	1000
	$\rho_0 = 0.2$															
0.00	0.08	−0.31	0.12	−0.14	39.75	21.10	15.95	11.35	4.57	1.42	1.20	0.49	40.56	21.28	16.06	11.42
0.25	0.04	−0.35	0.10	−0.14	39.73	21.10	15.94	11.34	4.53	1.39	1.19	0.49	40.54	21.27	16.04	11.41
0.50	−0.08	−0.48	0.07	−0.17	39.83	21.20	15.98	11.37	4.58	1.34	1.16	0.46	40.64	21.36	16.09	11.43
0.75	−0.55	−0.37	0.09	−0.10	41.42	21.82	16.18	11.50	4.27	1.52	1.19	0.54	42.08	21.93	16.23	11.53
0.95	−0.40	−0.45	0.08	−0.19	46.08	25.03	16.88	12.10	3.92	1.39	1.11	0.45	46.83	25.10	16.93	12.19
1.00	−0.37	−0.31	0.09	−0.20	49.58	27.41	17.79	13.19	3.31	1.12	0.92	0.30	50.60	27.58	17.90	13.24
	$\rho_0 = 0.5$															
0.00	0.63	−0.20	0.24	−0.11	47.05	24.76	18.68	13.27	6.11	1.91	1.54	0.64	48.26	25.04	18.84	13.37
0.25	0.61	−0.23	0.23	−0.11	47.03	24.76	18.66	13.27	6.08	1.88	1.53	0.64	48.21	25.03	18.82	13.36
0.50	0.47	−0.39	0.19	−0.14	47.25	24.97	18.74	13.32	6.14	1.83	1.49	0.61	48.46	25.22	18.89	13.40
0.75	0.06	−0.23	0.23	−0.05	50.30	26.20	19.10	13.58	5.92	2.10	1.58	0.72	51.33	26.40	19.18	13.62
0.95	0.29	−0.33	0.21	−0.19	58.05	31.50	20.25	14.66	5.54	2.05	1.51	0.64	59.15	31.67	20.35	14.77
1.00	−0.16	−0.36	0.13	−0.25	63.76	35.62	21.80	16.47	4.70	1.64	1.26	0.44	65.58	35.92	21.96	16.54
	$\rho_0 = 0.75$															
0.00	1.56	0.10	0.45	−0.01	49.25	25.63	19.28	13.71	7.70	2.43	1.86	0.80	50.72	25.98	19.48	13.80
0.25	1.57	0.08	0.44	−0.01	49.22	25.63	19.26	13.69	7.68	2.40	1.85	0.80	50.65	25.97	19.45	13.79
0.50	1.47	−0.09	0.40	−0.04	49.37	25.90	19.33	13.77	7.77	2.35	1.82	0.77	50.82	26.22	19.53	13.86
0.75	1.25	0.09	0.46	0.05	54.05	27.80	19.92	14.22	7.75	2.72	1.96	0.90	55.29	28.07	20.05	14.24
0.95	1.74	−0.08	0.38	−0.15	64.67	34.92	21.81	15.97	7.40	2.88	1.99	0.88	65.91	35.16	21.94	16.08
1.00	0.74	−0.49	0.03	−0.36	71.34	40.14	23.79	18.39	6.18	2.32	1.69	0.65	73.06	40.52	23.92	18.40

Notes: The true parameter value is $\beta_0 = 1$. See also the notes to Table 3a.

Table 4b
Size and power of the GMM and BMM estimators of β for the experiments with heteroskedastic non-Gaussian errors.

	GMM[b]								BMM							
	Size($\times 100$)				Power($\times 100$)				Size($\times 100$)				Power($\times 100$)			
δ	n															
	100	300	500	1000	100	300	500	1000	100	300	500	1000	100	300	500	1000
	$\rho_0 = 0.2$															
0.00	9.80	6.65	6.35	6.20	10.50	16.35	23.85	44.15	7.80	5.85	5.80	5.95	11.00	17.90	26.45	46.00
0.25	9.75	6.70	6.20	6.20	10.60	16.20	23.80	44.30	7.75	5.90	5.85	5.95	11.10	17.80	26.60	46.10
0.50	10.05	6.70	6.40	6.20	10.50	15.40	24.05	44.15	7.80	6.00	5.65	6.05	11.05	17.00	26.70	45.70
0.75	10.75	6.90	5.95	6.25	10.45	15.75	24.05	43.90	8.55	5.80	5.25	6.10	10.35	16.95	26.10	45.70
0.95	10.75	7.10	6.90	6.05	10.30	13.65	23.60	39.60	8.40	6.30	6.10	5.85	9.80	14.90	24.25	41.30
1.00	10.10	6.85	6.25	6.35	9.05	11.40	20.70	34.20	8.75	6.45	5.45	6.30	8.70	11.75	21.15	35.40
	$\rho_0 = 0.5$															
0.00	9.80	6.90	6.35	6.45	10.30	13.20	19.05	33.95	7.75	5.95	5.90	6.15	10.45	14.20	20.05	36.65
0.25	9.60	6.95	6.30	6.40	10.20	13.20	19.15	33.90	7.80	6.10	6.00	6.15	10.65	14.20	19.80	36.60
0.50	10.00	7.10	6.55	6.40	9.90	12.80	18.95	33.85	8.00	6.10	5.90	6.10	10.30	14.05	19.95	36.10
0.75	10.65	7.05	5.95	6.25	10.00	12.70	19.00	33.15	8.80	5.95	5.50	6.30	9.55	13.85	19.75	34.55
0.95	11.15	7.05	7.05	6.20	9.70	11.00	17.50	29.05	8.70	6.45	6.10	6.10	9.45	11.35	18.30	30.80
1.00	10.10	7.10	6.50	6.35	8.75	8.65	14.50	23.80	8.80	6.50	5.70	6.35	8.15	8.70	15.45	24.60
	$\rho_0 = 0.75$															
0.00	9.80	6.90	6.50	6.50	10.20	12.55	18.30	32.30	8.10	6.00	5.90	6.10	10.65	13.80	19.65	34.80
0.25	9.75	6.90	6.45	6.35	10.30	12.60	18.45	32.35	8.00	5.85	5.85	6.05	10.75	13.50	19.65	34.95
0.50	9.95	7.25	6.35	6.35	10.15	12.45	18.50	31.55	8.00	6.20	5.80	6.05	10.00	13.25	19.70	34.30
0.75	11.10	6.95	6.15	6.40	10.05	11.90	18.10	30.80	9.05	5.85	5.70	6.15	9.55	13.20	18.90	32.10
0.95	10.95	7.35	7.00	6.30	9.60	10.30	15.85	25.70	9.10	6.80	6.40	6.05	9.90	10.45	17.10	27.35
1.00	9.80	7.40	6.70	6.60	8.65	7.90	12.50	19.85	8.80	6.70	5.75	6.45	8.50	8.75	14.10	21.25

Notes: The true parameter value is $\beta_0 = 1$ and power is calculated at 0.8. See also the notes to Table 3a.

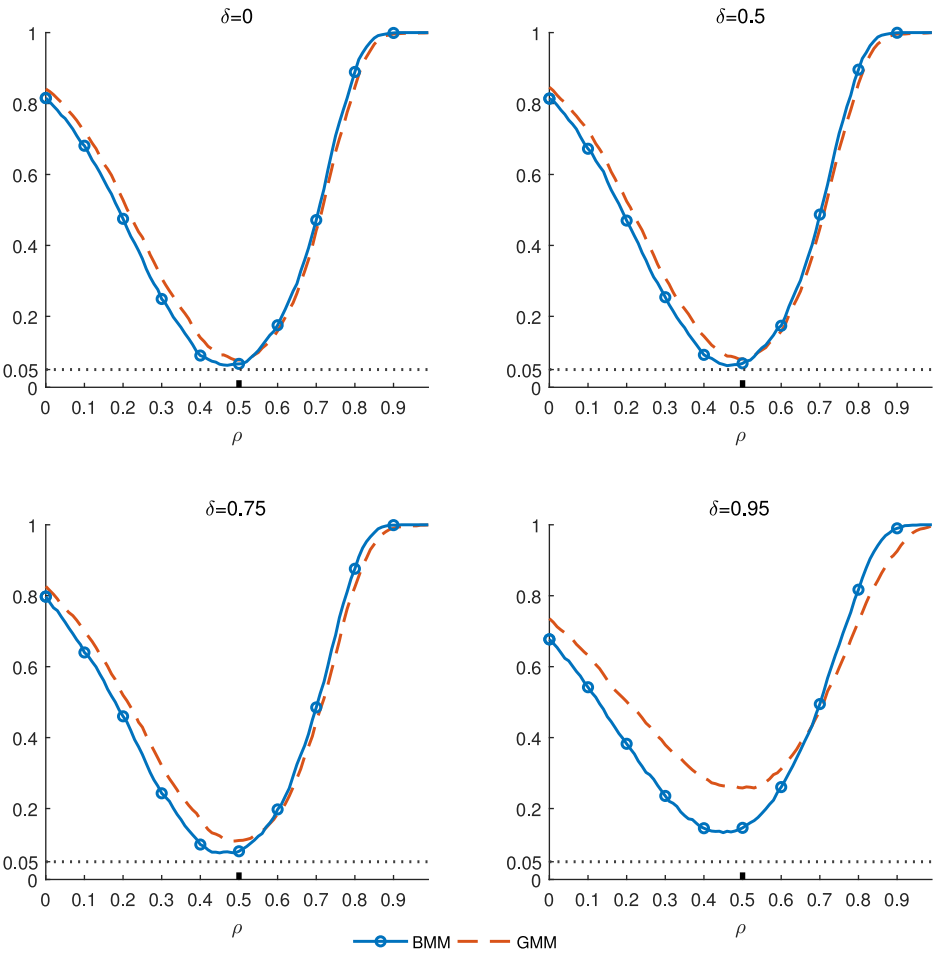


Fig. 1a. Empirical power functions for ρ in the case of $\rho_0 = 0.5$, $n = 100$, and homoskedastic non-Gaussian errors for different values of δ.

We now turn to size and power properties of the BMM and GMM estimators. As can be seen from Table 1b, overall the tests of ρ have empirical size close to the nominal size of 5% when $\delta \leq 0.75$. This is true for both estimators. When the sample size is small ($n = 100$), the GMM estimator slightly over-rejects the null if the degree of spatial autocorrelation is high ($\rho_0 = 0.75$), and the size distortion becomes more severe as ρ_0 is increased towards unity. In comparison, the BMM estimator has the correct empirical size even when the sample size is small and ρ_0 is close to unity when $\delta \leq 0.75$. As the sample size becomes larger ($n \geq 300$), both estimators have the correct size and reasonable power for all values of ρ_0 if $\delta \leq 0.75$. These results suggest that the condition $\delta < 1/2$ assumed in the paper might be too conservative, and whilst sufficient it might not be necessary. Turning to size and power of the tests for β summarized in Table 2b, we note that both estimators perform well, yielding the correct size and high power, and their performance is overall better as compared to the results we obtain for ρ. These findings seem to be quite robust to non-Gaussian errors.[20] Finally, both estimators display larger size distortion under heteroskedasticity than under homoskedasticity when $n = 100$ and $\delta \leq 0.75$, but they become correctly sized when n increases to 300 and $\delta \leq 0.75$. As in the case of homoskedasticity, the GMM estimator is more severely over-sized than the BMM estimator under heteroskedasticity if δ is close to unity and the sample size is small. *Overall, our extensive simulation results suggest that in practice one can perform sound inference with the BMM estimator so long as δ is not too close to unity.*

Figs. 1a and 1b display the power functions for ρ under homoskedasticity when $\rho_0 = 0.5$ for $n = 100$ and 300, respectively. Overall, the tests of $\rho = \rho_0$ based on GMM and BMM estimators have similarly good power when $\delta \leq 0.5$. As δ moves towards one, the tests based on both estimators tend to over-reject the null. The over-rejection is more severe for the GMM estimator than the BMM estimator. For example, as shown in Fig. 1a, when $\delta = 0.95$ and $n = 100$ the rejection frequency of the GMM estimator under the null is 25.8% as compared to 14.6% for the BMM estimator. A comparison of Figs. 1a and 1b reveals that when $\delta \leq 0.75$ the size distortion is reduced as n expands from 100 to 300,

[20] The size and power properties of both estimators under homoskedastic Gaussian errors can be found in Tables S.7–S.10 of the Online Supplement.

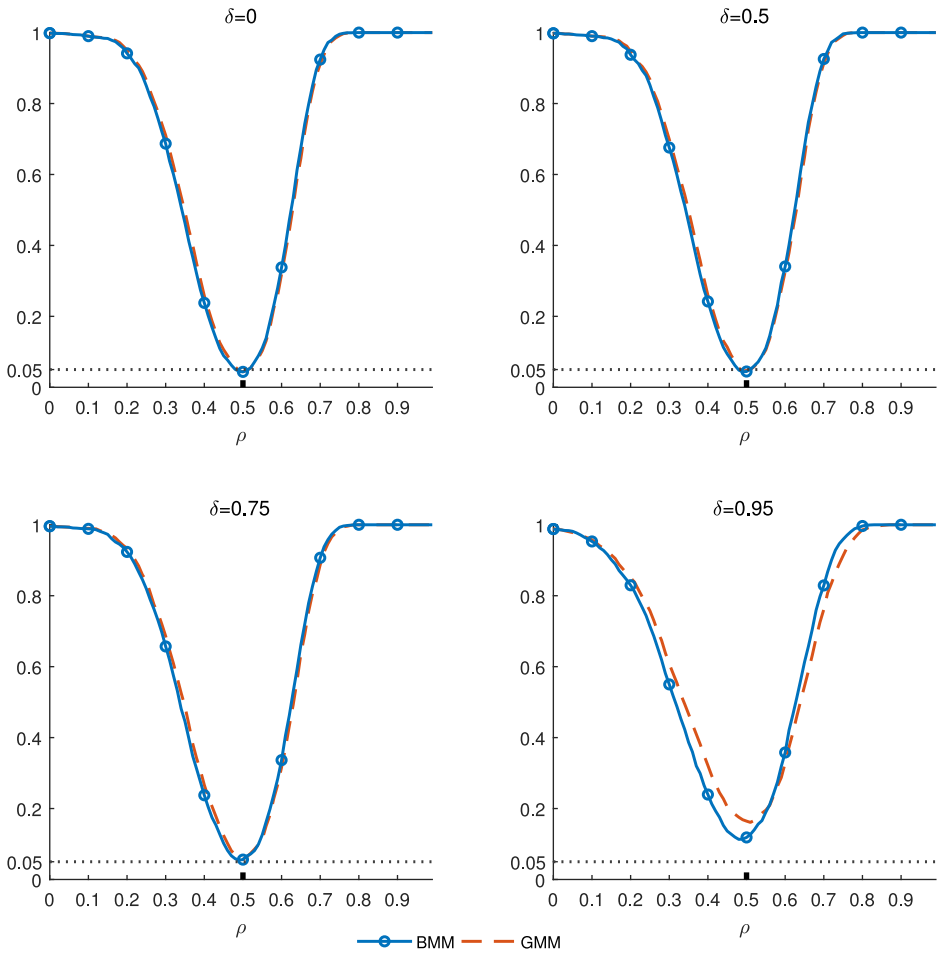


Fig. 1b. Empirical power functions for ρ in the case of $\rho_0 = 0.5$, $n = 300$, and homoskedastic non-Gaussian errors for different values of δ.

but the over-rejection does not disappear with increasing sample size when $\delta = 0.95$. These findings are in line with our theoretical results.[21] We proceed with Figs. 2a and 2b, which show the power functions for β when $\beta_0 = 1$ for $n = 100$ and 300, respectively. We see at once that the power curves for both estimators are very close. We also note that the over-rejection is less of a problem for the estimators of β than for ρ. The power is relatively low when $n = 100$ but rises notably as n increases to 300. The power functions under heteroskedasticity resemble the plots under homoskedasticity with slightly larger size distortion when the sample size is small. See the Online Supplement for additional empirical power functions under heteroskedasticity.

7. Empirical application to US sectoral prices

In earlier studies using US input–output tables Acemoglu et al. (2012) and Pesaran and Yang (2020) find that δ, the degree of centrality of the US production network, lies between 0.72 and 0.82, and accordingly the standard assumption in the spatial econometrics literature that presumes all units are non-dominant is violated. In what follows we first extend the closed economy multi-sectoral model in Pesaran and Yang (2020) to a small open economy in which production also requires imported intermediate inputs (raw materials). We then apply the GMM and BMM estimation techniques to investigate the degree of interdependence in sectoral price changes in the US economy.

For simplicity, we assume that there is only one type of imported intermediate good, whose quantity demanded for production by sector i at time t is denoted by m_{it}. Each sector i at time t produces output, q_{it}, by the following

[21] Similar findings hold for different values of ρ_0 whether the errors are Gaussian or non-Gaussian, as can be seen from the power plots in the Online Supplement.

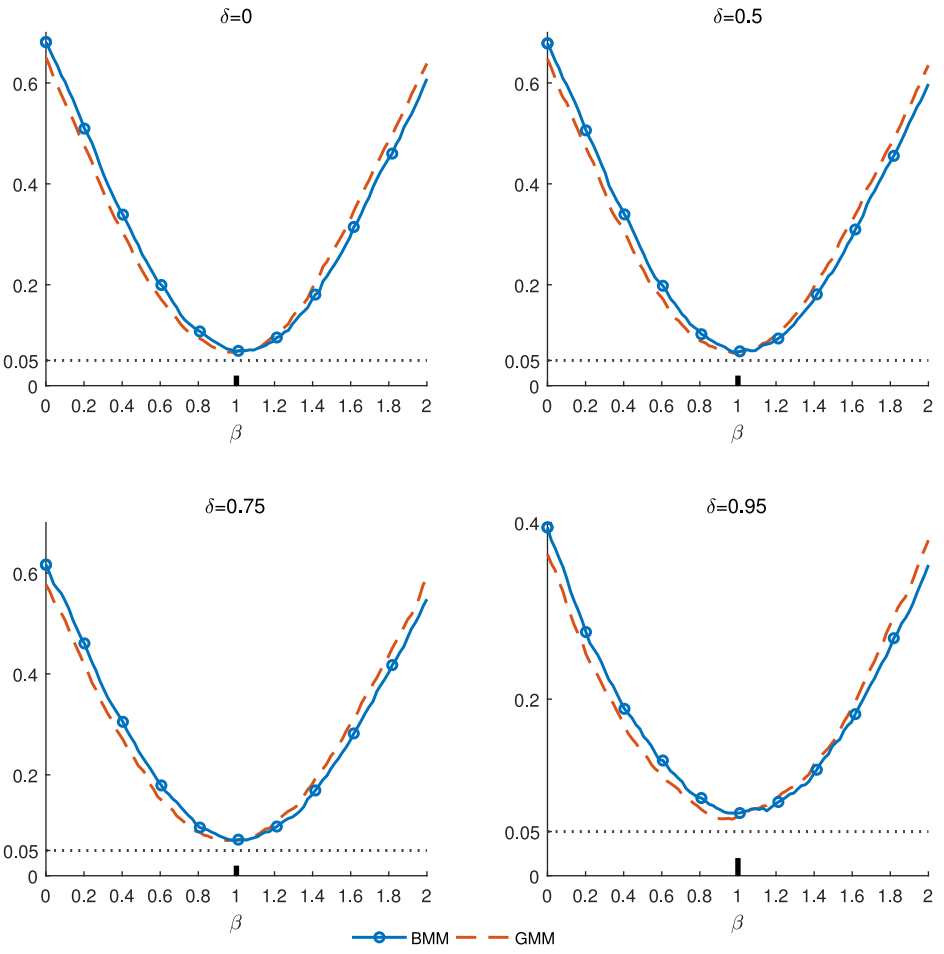


Fig. 2a. Empirical power functions for β in the case of $\beta_0 = 1$, $n = 100$, and homoskedastic non-Gaussian errors for different values of δ.

Cobb–Douglas production technology:

$$q_{it} = e^{\alpha u_{it}} l_{it}^{\alpha} m_{it}^{\vartheta} \prod_{j=1}^{n} q_{ij,t}^{(1-\alpha-\vartheta)w_{ij}}, \quad \text{for } i = 1, 2, \ldots, n, \tag{63}$$

where l_{it} is the labor input, $q_{ij,t}$ is the amount of output of sector j used by sector i, and $u_{it} = \gamma_i f_t + v_{it}$ is the productivity shock that consists of two components: v_{it}, which is a sector-specific shock, and f_t, which is a common factor with heterogeneous factor loadings, γ_i, for $i = 1, 2, \ldots, n$. The parameter α represents the share of labor, ϑ represents the share of imported intermediate goods, and w_{ij} is the share of sector j's output in the total domestic intermediate input use by sector i.

The representative household is assumed to have Cobb–Douglas preferences over n goods:

$$u\left(c_{1t}, c_{2t}, \ldots, c_{nt}\right) = A \prod_{i=1}^{n} c_{it}^{1/n}, A > 0. \tag{64}$$

where c_{it} is the quantity consumed of good i. Furthermore, the household is endowed with l_t unit of labor, supplied inelastically at wage rate Wage_t. In equilibrium, the commodity markets clear,

$$c_{it} = q_{it} - \sum_{j=1}^{n} q_{ji,t} - q_{x,it}, \quad \text{for } i = 1, 2, \ldots, n,$$

where $q_{x,it}$ is the quantity exported of good i; the labor market clears, $l_t = \sum_{i=1}^{n} l_{it}$; and trade is balanced, $P_{m,t} \sum_{i=1}^{n} m_{it} = \sum_{i=1}^{n} P_{it} q_{x,it}$, where P_{it} denotes the price of good i, and $P_{m,t}$ denotes the exogenous world price of the imported intermediate good.

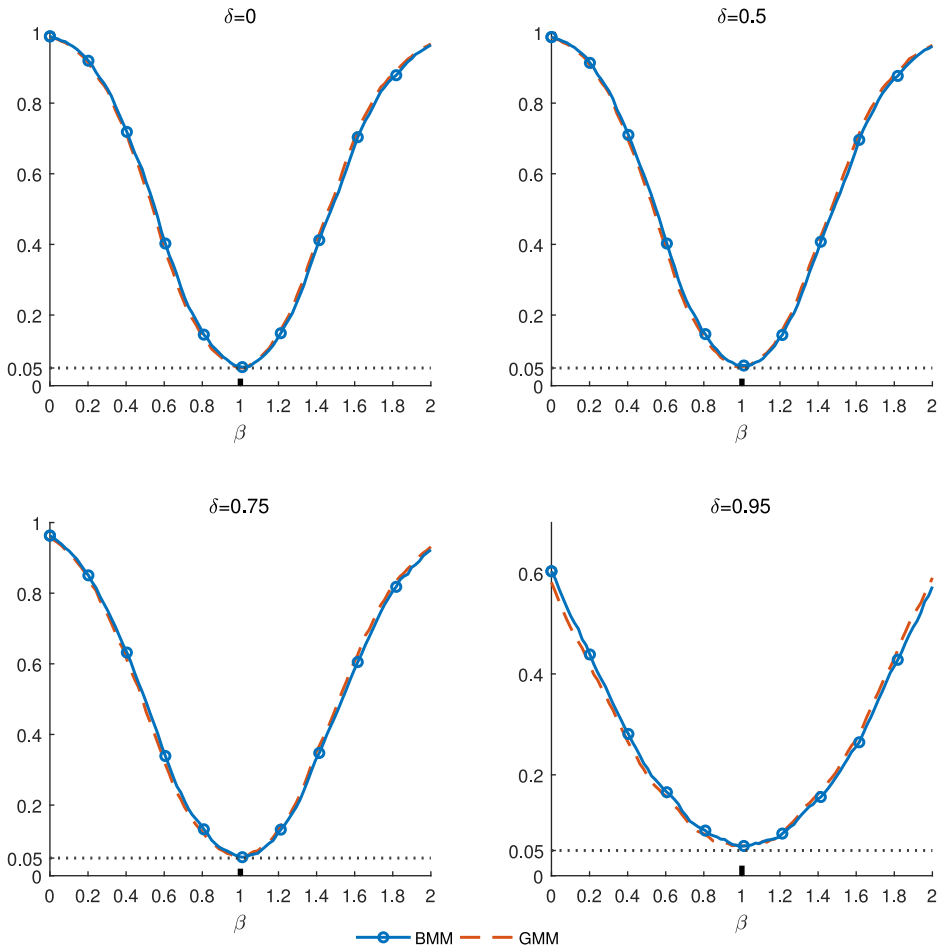


Fig. 2b. Empirical power functions for β in the case of $\beta_0 = 1$, $n = 300$, and homoskedastic non-Gaussian errors for different values of δ.

Given prices $\{P_{1t}, P_{2t}, \ldots, P_{nt}, P_{m,t}, \text{Wage}_t\}$, the profit-maximization problem of sector i, for $i = 1, 2, \ldots, n$, is given by

$$\max_{q_{ij,t}, l_{it}, m_{it}} \left\{ P_{it} e^{\alpha u_{it}} l_{it}^{\alpha} m_{it}^{\vartheta} \prod_{j=1}^{n} q_{ij,t}^{(1-\alpha-\vartheta)w_{ij}} - \text{Wage}_t \times l_{it} - P_{m,t} m_{it} - \sum_{j=1}^{n} P_{jt} q_{ij,t} \right\}.$$

The first-order conditions with respect to $q_{ij,t}$, l_{it}, and m_{it} imply that

$$q_{ij,t} = \frac{(1-\alpha-\vartheta)\, w_{ij} P_{it} q_{it}}{P_{jt}}, \quad l_{it} = \frac{\alpha P_{it} q_{it}}{\text{Wage}_t}, \quad m_{it} = \frac{\vartheta P_{it} q_{it}}{P_{m,t}}. \tag{65}$$

Substituting (65) into (63) and after some simplifications yields

$$p_{it} = \rho \sum_{j=1}^{n} w_{ij} p_{jt} + \alpha \omega_t + \vartheta p_{m,t} - b_i - \alpha\left(\gamma_i f_t + v_{it}\right), \text{ for } i = 1, 2, \ldots, n, \tag{66}$$

where $\rho = 1-\alpha-\vartheta$, $p_{it} = \log(P_{it})$, $\omega_t = \log(\text{Wage}_t)$, $p_{m,t} = \log\left(P_{m,t}\right)$, and $b_i = \alpha \log(\alpha) + \vartheta \log(\vartheta) + \rho \log(1-\alpha-\vartheta) + \rho \sum_{j=1}^{n} w_{ij} \log(w_{ij})$.

The system of price equations in (66) is in the form of a panel SAR model with fixed effects, observed (ω_t and $p_{m,t}$) and unobserved (f_t) common factors. To transform these equations into a SAR model in observables we take first differences[22]

[22] This paper focuses on cross section SAR models. The estimation of the panel data model given by (66) is beyond the scope of the current paper.

$$\Delta p_{it} = \rho \sum_{j=1}^{n} w_{ij} \Delta p_{jt} + \alpha \Delta \omega_t + \vartheta \Delta p_{m,t} - \alpha \left(\gamma_i \Delta f_t + \Delta v_{it} \right), \text{ for } t = 1, 2, \ldots, T, \tag{67}$$

and consider time averages computed over the sample period $t = 1, 2, \ldots, T$ to obtain

$$\overline{\Delta p}_i = \rho \sum_{j=1}^{n} w_{ij} \overline{\Delta p}_j + \alpha \overline{\Delta \omega} + \vartheta \overline{\Delta p}_m - \alpha \left(\gamma_i \overline{\Delta f} + \overline{\Delta v}_i \right), \tag{68}$$

where $\overline{\Delta p}_i = \frac{1}{T} \sum_{t=1}^{T} \Delta p_{it}$, $\overline{\Delta \omega} = \frac{1}{T} \sum_{t=1}^{T} \Delta \omega_t$, $\overline{\Delta p}_m = \frac{1}{T} \sum_{t=1}^{T} \Delta p_{m,t}$, $\overline{\Delta f} = \frac{1}{T} \sum_{t=1}^{T} \Delta f_t$, and $\overline{\Delta v}_i = \frac{1}{T} \sum_{t=1}^{T} \Delta v_{it}$. For a given sample period $\overline{\Delta \omega}$, $\overline{\Delta p}_m$, and $\overline{\Delta f}$ are fixed, and only cross section variations are relevant for the estimation of ρ. We also assume that the factor loadings follow the random coefficient model $\gamma_i = \gamma_0 + \eta_i$, where $\eta_i \sim IID(0, \sigma_\eta^2)$, for $i = 1, 2, \ldots, n$. Using this result in (68), we now have

$$\overline{\Delta p}_i = a + \rho \sum_{j=1}^{n} w_{ij} \overline{\Delta p}_j + \varepsilon_i, \tag{69}$$

where $a = \alpha \overline{\Delta \omega} + \vartheta \overline{\Delta p}_m - \alpha \gamma_0 \overline{\Delta f}$, and $\varepsilon_i = -\alpha \left(\overline{\Delta v}_i + \overline{\Delta f} \eta_i \right)$. The SAR model in the rate of price changes, (69), can now be estimated by GMM and BMM. The parameter of interest is the spatial coefficient, ρ, which can be interpreted as capital's share of output. The $n \times n$ matrix $\mathbf{W} = \left(w_{ij} \right)$ that summarizes the input–output relations corresponds to the spatial weights matrix.

The spatial weights matrix, $\mathbf{W}$, is constructed from the input–output tables at the most disaggregated level obtained from the website of the Bureau of Economic Analysis (BEA). These tables cover around 400 industries and are compiled by the BEA every five years. Specifically, $\mathbf{W}$ is a commodity-by-commodity direct requirements matrix, of which the (i, j)th entry represents the expense on commodity j per dollar of production of commodity i.[23] The commodity-by-commodity direct requirements (**DR**) tables are derived from the commodity-by-commodity total requirements (**TR**) tables by the following formula: $\mathbf{DR} = (\mathbf{TR} - \mathbf{I}) (\mathbf{TR})^{-1}$, where $\mathbf{I}$ is the identity matrix of conformable dimension. The $\mathbf{W}$ matrix is taken as the transpose of **DR** and standardized so that the sum of intermediate input shares (the row sum of $\mathbf{W}$) equals unity for every sector. Since the vast majority of the elements in $\mathbf{W}$ are rather small numbers, in order to reduce noise in the system we construct a robust weights matrix by setting each element of $\mathbf{W}$ to one if it is greater than or equal to a given threshold value ϵ_w $(0 < \epsilon_w < 1)$, and to zero otherwise. Then the sectors with zero row sums are dropped and the matrix is row-standardized so that each row sums to one. The resulting matrix is denoted by $\tilde{\mathbf{W}}(\epsilon_w)$. The sector-specific price index at annual frequency is obtained from the BEA's gross domestic product by industry accounts. The annual rates of price changes are computed over the period 1998–2015, and they are matched to the sectors in the input–output tables using the BEA industry codes.

Given the time range of the price data, we consider two versions of $\tilde{\mathbf{W}}$ constructed from the input–output tables for the years 2002 and 2007, denoted by $\tilde{\mathbf{W}}_{2002}$ and $\tilde{\mathbf{W}}_{2007}$, respectively. In particular, we consider a cut-off value $\epsilon_w = 10\%$, which means that for any given sector only important suppliers that contribute at least 10% of the total input purchases are taken into account.[24]

We begin by examining the δ-dominance of the production networks for the years 2002 and 2007 by applying the extremum estimator developed by Pesaran and Yang (2020) to the outdegrees of the filtered input–output matrices $\tilde{\mathbf{W}}_{2002}(0.1)$ and $\tilde{\mathbf{W}}_{2007}(0.1)$. Table 5 reports the estimates of δ for the top five most important sectors for these weights matrices. The results show that the highest degree of dominance, $\widehat{\delta}_{(1)}$, lies between 0.71 and 0.85, and are not close to unity. Therefore, our proof of consistency of the GMM and BMM estimators of the spatial parameter applies to this empirical application. But for valid inference our proofs require $\delta < 1/2$, and special care must be exercised when carrying out inference on ρ in the present empirical application. Although, as noted above, our Monte Carlo experiments suggest that the degree of over-rejection of tests based on the BMM estimator of ρ is relatively low so long as δ is not too close to unity, and inference based on the BMM estimators seems to be acceptable for values of δ around 0.75.

Turning to the sectoral price changes, to allow for the possibility of structural breaks due to the 2007–2008 financial crisis, we consider two sub-samples: the pre-financial crises (1998–2006) and the post-financial crises (2007–2015) periods. The weights matrix $\tilde{\mathbf{W}}_{2002}(0.1)$ is used for the first sub-sample, while $\tilde{\mathbf{W}}_{2007}(0.1)$ is used for the second sub-sample. The BMM estimates are computed by (38) and (48) if the errors are assumed to be homoskedastic and heteroskedastic, respectively. The GMM estimates refer to the best GMM and are obtained in two steps: In the first step, we compute initial consistent estimate, $\tilde{\rho}$, by (27) using two equally weighted quadratic moments with $\mathbf{B}_\ell = \tilde{\mathbf{W}}^\ell - n^{-1} Tr\left(\tilde{\mathbf{W}}^\ell\right) \mathbf{I}_n$, $\ell = 1, 2$, if the errors are assumed to be homoskedastic, or $\mathbf{B}_\ell = \tilde{\mathbf{W}}^\ell - Diag\left(\tilde{\mathbf{W}}^\ell\right)$, $\ell = 1, 2$, assuming

[23] The words commodity and sector are used interchangeably to convey the same meaning throughout this section.

[24] Our choice of the 10% threshold for non-zero elements of the weights matrix is in line with the US Regulation SFAS No. 131 that requires public firms to report customers representing more than 10% of their total yearly sales (see Cohen and Frazzini, 2008, p. 1978). The results for other cut-off values of $\epsilon_w = 5\%$ and 7.5% are provided in the Online Supplement. Using lower threshold values tend to yield higher estimates of ρ.

Table 5
Estimates of the degree of dominance, δ, of the top five pervasive sectors using US input–output tables.

	Input–output table for 2002		Input–output table for 2007	
	$\mathbf{W}_{2002}$	$\tilde{\mathbf{W}}_{2002}\,(0.1)$	$\mathbf{W}_{2007}$	$\tilde{\mathbf{W}}_{2007}\,(0.1)$
$\widehat{\delta}_{(1)}$	0.778	0.851	0.724	0.705
$\widehat{\delta}_{(2)}$	0.759	0.796	0.651	0.703
$\widehat{\delta}_{(3)}$	0.597	0.642	0.608	0.695
$\widehat{\delta}_{(4)}$	0.550	0.422	0.592	0.565
$\widehat{\delta}_{(5)}$	0.546	0.402	0.553	0.491
n	313 [301]	286 [114]	384 [364]	350 [140]
n^*	69,268 (70.70%)	581 (0.71%)	107,619 (72.98%)	616 (0.50%)

Notes: $\tilde{\mathbf{W}}(\epsilon_w = 0.1)$ denotes a filtered version of $\mathbf{W} = (w_{ij})$, defined by $\tilde{\mathbf{W}}(\epsilon_w) = \left(\tilde{w}_{ij}(\epsilon_w)\right)$, where $\tilde{w}_{ij}(\epsilon_w)$ is a row-standardized version of $w^*_{ij}(\epsilon_w)$ defined by $w^*_{ij}(\epsilon_w) = w_{ij} I\left(w_{ij} \geq \epsilon_w\right)$, where $I(A)$ is an indicator variable which takes the value of unity if A holds and zero otherwise. We set $\epsilon_w = 10\%$ and report $\hat{\delta}_{(1)} > \hat{\delta}_{(2)} > \cdots > \hat{\delta}_{(5)}$, the five largest estimates of δ corresponding to the outdegrees of $\mathbf{W}$ and $\tilde{\mathbf{W}}(0.1)$, for the years 2002 and 2007. n is the total number of sectors with non-zero total demands (indegrees). The numbers in square brackets are the numbers of sectors with non-zero outdegrees. Note that a few sectors were dropped when constructing $\tilde{\mathbf{W}}$ from $\mathbf{W}$, since their total demands become zero. n^* is the number of non-zero elements. The percentages of non-zero elements are in parentheses.

Table 6
Estimation results of the cross-section model (69).

Year	Sub-sample 1998–2006		Sub-sample 2007–2015	
	BMM	GMM[b]	BMM	GMM[b]
	Assuming homoskedastic errors			
$\hat{\rho}$ [Share of capital]	0.397†	0.396†	0.287†	0.281†
	(0.106)	(0.106)	(0.072)	(0.073)
$\hat{\sigma}^2_\eta$ [Error variance]	7.810	7.813	2.586	2.592
R^2	0.219	0.218	0.159	0.154
	Assuming heteroskedastic errors			
$\hat{\rho}$ [Share of capital]	0.542†	0.542†	0.241†	0.246†
	(0.115)	(0.114)	(0.069)	(0.069)
Weights matrix	$\tilde{\mathbf{W}}_{2002}\,(0.1)$		$\tilde{\mathbf{W}}_{2007}\,(0.1)$	
n [Number of sectors]	286		350	

Notes: All estimations include an intercept (not shown here). Standard errors are in parentheses. † indicates significance at 1% level. The spatial weights matrices are constructed with a threshold value of $\epsilon_w = 10\%$. $\tilde{\mathbf{W}}_{2002}\,(0.1)$ is used in the estimation over the period 1998–2006; $\tilde{\mathbf{W}}_{2007}\,(0.1)$ is used in the estimation over the period 2007–2015. R^2 is computed by (59) assuming homoskedasticity. The BMM estimates assuming homoskedastic errors are computed by (38), and computed by (48) if assuming heteroskedastic errors.
[b] The GMM estimator refers to the best GMM estimator computed by a two-step procedure following (27) using the $\tilde{\mathbf{G}} - n^{-1}Tr\left(\tilde{\mathbf{G}}\right)\mathbf{I}_n$ if the errors are assumed to be homoskedastic, and $\tilde{\mathbf{G}} - Diag\left(\tilde{\mathbf{G}}\right)$ if assuming heteroskedasticity, where $\tilde{\mathbf{G}} = \mathbf{G}(\tilde{\rho})$ is evaluated at the first-step estimate, $\tilde{\rho}$.

error heteroskedasticity.[25] In the second step, we re-estimate the model using $\tilde{\mathbf{G}} - n^{-1}Tr\left(\tilde{\mathbf{G}}\right)\mathbf{I}_n$ under homoskedasticity and $\tilde{\mathbf{G}} - Diag\left(\tilde{\mathbf{G}}\right)$ under heteroskedasticity, where $\tilde{\mathbf{G}} = \mathbf{G}(\tilde{\rho})$ is evaluated at the first-step estimate.

Table 6 presents the estimation results of model (69). The top panel gives the estimates obtained under homoskedasticity and the bottom panel gives the estimates that allow for heteroskedasticity. As can be seen, the BMM and GMM estimates are very close to one another, irrespective of whether the errors are assumed to be homoskedastic or not. Under homoskedastic errors, the estimated share of capital is around 0.40 for the first sub-sample and 0.29 for the second sub-sample. When we allow for heteroskedasticity we obtain a much larger estimate of 0.54 for the first sub-sample, and a slightly smaller estimate of 0.25 for the second sub-sample, which suggest heteroskedasticity is likely to be more severe in the first as compared to the second sub-sample. Despite the wide range of estimates of ρ obtained over the two sub-periods (0.28–0.54), they match reasonably well with the commonly documented values of share of capital in the literature. The most commonly used value in calibration exercises is 0.36 (Hansen and Wright, 1998 ; Danthine et al., 2008). Other frequently used calibration values fall in the range 0.3–0.4. For example, Cooley and Prescott (1995) suggest 0.4; Gollin (2002) recommends a range of 0.23–0.34; Danthine et al. (2008) uses 0.3.

[25] Here we write $\tilde{\mathbf{W}}_{2002}\,(0.1)$ and $\tilde{\mathbf{W}}_{2007}\,(0.1)$ as $\tilde{\mathbf{W}}$ to simplify the notations.

8. Concluding remarks

An important assumption in the spatial econometrics literature requires that the weights (connections) matrix is uniformly bounded in both row and column sums. This assumption excludes the existence of dominant units in the network and is too restrictive for many economic applications. The current paper relaxes this assumption and allows the centrality of the connections to rise at the rate of δ with n, as compared to the value of $\delta = 0$ assumed in the literature. We also establish the asymptotic distribution of the GMM estimator due to Lee (2007) and Lin and Lee (2010) under this more general setting, and propose a new BMM estimator which is simple to compute and has better small sample properties as compared to the best GMM estimator when the degree of centrality of the weights matrix, δ, is relatively large. Asymptotic properties of both estimators are investigated under homoskedastic as well as heteroskedastic errors, and shown to be consistent and normally distributed if the maximum absolute column sum of the interaction matrix does not increase too fast as n grows. For consistent estimation it is required that $\delta < 1$, and for the validity of the asymptotic distribution we need $\delta < 1/2$. But the extensive Monte Carlo experiments reported in the paper and in the Online Supplement suggest that GMM and BMM estimators could perform reasonably well if $\delta \leq 0.75$. Thus, it might be conjectured that the sufficient condition of $\delta < 1/2$ might not be necessary for the validity of the asymptotic distribution of GMM and BMM estimators. Further analysis is required if $\delta > 1/2$. Such an analysis is not covered in the present paper.

Note: Supplementary material providing proofs, additional Monte Carlo and empirical results related to this article can be found online at https://doi.org/10.1016/j.jeconom.2020.04.045. Replication files can also be accessed online at https://doi.org/10.1016/j.jeconom.2020.04.045.

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